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Edge-Deployed Multimodal LLMs for Wildfire Decision Support in Critical Operations

Bachelor of Science Thesis in Software Engineering and Management

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Abstract—Wildfire reconnaissance requires rapid decision-making under conditions that frequently preclude network connectivity, yet existing detection systems offer no operational guidance. This thesis designs and evaluates an edge-deployable pipeline combining YOLOv8-based fire detection with multimodal large language models (MLLMs) reasoning over pre-loaded geographic context to produce structured tactical decision support offline on consumer-grade hardware. Three architectures (Qwen3-VL 4B, Ministral 3B, and Gemma 4 E2B) were evaluated through quantitative benchmarking and qualitative interviews with wildfire response professionals across two Design Science Research cycles. Model performance shifted substantially between cycles as prompt and context complexity increased, with Qwen3-VL achieving strong classification accuracy on valid outputs but failing frequently under production conditions, Ministral producing the most complete decision support but suffering severe context bias, and Gemma remaining reliable but shallow. Practitioners confirmed that geographically grounded recommendations aligned with their operational assessment process. The results establish feasibility of offline MLLM decision support on consumer hardware, while showing that richer prompts and geographic context can degrade classification reliability in ways that prompt engineering alone cannot resolve.

Index Terms—Decision Support, Design Science Research, Edge Deployment, Multimodal LLMs, Wildfire Detection

I. INTRODUCTION

Climate change and its related extreme weather events are increasing in intensity globally, and the severity of wildfires in particular is escalating at an alarming rate, with extreme wildfires doubling from 2003–2023 [1], [2]. The role of software in terms of efficient detection and response is vital in ensuring the consequences of wildfires are minimized. Research in this area is important and aligns with The United Nations Sustainable Development Goals, specifically SDG 13: “Take urgent action to combat climate change and its impacts.” The consequences of insufficient decision-support in high-stakes wildfire management are severe and impact wildlife, human lives, and infrastructure. Research in human–AI collaboration shows that explainable AI systems improve task performance and reduce cognitive load in critical decision-making [3], [4]. To address these needs, the Swedish Defence Materiel Administration (FMV) serves as the stakeholder for this research, together with other organizations that rely on

critical decision-making systems such as emergency services in field operations.

Effective wildfire reconnaissance requires rapid and reliable decision-making with the help of aerial imagery, particularly in contexts where stakes are high and resources are limited. Traditional detection systems based on deep-learning object detectors achieve satisfactory accuracy in localizing fires and smoke but do not provide useful guidance for human decision-makers, as their raw output lacks explanations or tactical recommendations [5]. The need for more easily interpreted AI outputs is important to reduce cognitive load for the user [6].

Furthermore, in many operational environments, reliable connectivity and network bandwidth are often limited. To address this, edge-deployment, i.e. running models locally, is crucial. However, wildfire detection systems and multimodal large language models (MLLMs) have to date been developed along largely independent research lines. On-device use of MLLMs to augment traditional object detection with natural language decision support remains, to our knowledge, unexplored. This gap is particularly critical in emergency response contexts where explainable AI outputs directly impact outcomes.

To address these challenges, this thesis designs and evaluates an edge-deployable hybrid system that augments traditional object detection with MLLMs for natural language decision support in wildfire reconnaissance. Three architectures, Qwen3-VL:4b, Gemma4:e2b, and Ministral3:3b, representing different geographic origins and technical designs, are deployed and benchmarked on identical consumer-grade hardware. Specifically, we investigate the following research questions:

- RQ1:** How can lightweight MLLMs complement traditional object detection models in filtering, interpreting, and explaining visual detection results for tactical decision support?
- RQ2:** What trade-offs exist between different MLLM architectures in terms of detection accuracy, inference speed, hardware requirements, and cognitive support effectiveness?

We follow Design Science Research methodology across two cycles, conducting mixed-methods evaluation through quantitative benchmarking and qualitative stakeholder interviews to assess decision support quality.

II. BACKGROUND

A. Wildfire Detection in Critical Operations

Wildfire reconnaissance involves monitoring and detecting active fires to support decision-making for emergency response personnel under time pressure and in areas with limited connectivity, creating a need for systems that can run offline on consumer hardware.

Reconnaissance can draw on several data sources, including aerial spotters and UAVs deployed on scene [7], [8]. UAVs equipped with infrared cameras are preferred for identifying hotspots and tracking fire spread [8]. This thesis focuses on standard RGB drone imagery as the most widely available input format.

Analysing incoming imagery is a continuous task subject to human limitations such as fatigue and reduced attention over time [9]. Responders must simultaneously assess many factors including natural fire breaks, vegetation type, wind, terrain, water access, and proximity of structures that may require evacuation [7].

B. Object Detection with YOLO

YOLO is a single-stage object detection architecture that simultaneously localizes and classifies objects in an image in a single forward pass, making it well suited for real-time applications [10]. This thesis uses a pre-trained YOLOv8 model fine-tuned for forest fire and smoke detection [11], producing bounding boxes for two classes: Fire and Smoke. Each detection includes a class label and a confidence score. A key strength of the architecture is that it can be trained on domain-specific datasets, allowing models to specialize in detecting particular objects of interest.

C. Multimodal Large Language Models

Multimodal Large Language Models (MLLMs) combine visual and textual understanding, enabling AI systems to process images and generate natural language responses that go beyond simple object detection [12]. MLLMs learn through instruction tuning, allowing them to handle different tasks without task-specific retraining [13].

The architectures evaluated in this research come in different parameter count variants. Parameters are the internal values a model learns during training, and their number largely determines how much the model can learn, how fast it runs, and how much memory and compute it requires at inference time.

1) *Minstral 3 (Mistral AI)*: Minstral 3 is a family of dense multimodal models developed by Mistral AI (France), representing European open-weight LLM development [14]. This research evaluates the 3B parameter variant.

2) *Qwen3-VL (Alibaba)*: Qwen3-VL is the vision-language variant of Alibaba's Qwen3 family, representing Chinese MLLM development [15]. It utilizes dynamic resolution for its vision encoder to handle varying input scales. This research evaluates the 4B dense variant.

3) *Gemma 4 (Google)*: Gemma 4 is Google DeepMind's open-weight multimodal model family designed for edge deployment [16]. This research evaluates the E2B variant, which utilizes a Per-Layer Embedding (PLE) architecture.

D. Edge AI and Resource Constraints

LLMs traditionally run on centralized cloud servers in massive data centers requiring large amounts of energy and hardware resources. For certain use cases, however, such an arrangement is simply not feasible. Hence, there exists use cases for edge-deployed AI, where inference is done locally on the device. In crisis situations, this offers critical advantages: low latency, data privacy and offline functionality.

In this work, the edge environment is primarily consumer-grade laptops, which are built with general-purpose central processing units (CPU) and utilize considerably less powerful graphics processing units (GPU) compared to server-grade equivalents. A laptop's random access memory (RAM) is typically limited to 16–32 gigabytes, while frontier LLMs can require tens to hundreds of gigabytes of RAM, far exceeding these limits.

Optimizations of edge-deployed LLMs result in tradeoffs: reduced memory but improved speed comes at the cost of accuracy. Therefore, selecting a balance is key to effectively make use of edge-deployed models.

E. Decision Support Systems

A decision support system (DSS) is a computer-based tool that assists human operators in making decisions by analyzing available information and presenting it in a form that supports faster, more informed judgement [17], [18]. DSSs are applied across domains including healthcare, manufacturing, emergency response, and military operations, and their defining characteristic is that they inform rather than replace the operator's decision [18].

In high-stakes operational contexts, DSSs play a particular role: converting available information into a form that operators can act on under time pressure and uncertainty [17]. A common framing distinguishes decision support from automation: an automated system acts on the operator's behalf, whereas a DSS structures information so the operator can decide. The operator retains responsibility for the final action, which is essential in domains where the cost of an incorrect automated decision is high.

III. RELATED WORK

A. Multimodal Vision-Language Models

Mai et al. provide an extensive survey of the field, highlighting that multimodal models enable AI systems to process and understand several data types simultaneously, much like humans do [19]. LLaVA introduced instruction tuning for multimodal models using GPT-4-generated training data, achieving 85.1% performance relative to GPT-4 on visual understanding tasks [20]. Meanwhile, Qwen-VL demonstrated strong capabilities across multiple vision-language tasks, including visual understanding, optical character recognition,

and visual question answering [21]. Recent work on deploying these models on edge devices has shown that smaller multimodal LLMs can operate efficiently on consumer hardware while remaining competitive compared to larger variants deployed on the cloud [22]. These models are excellent at visual understanding and question answering but have not yet been systematically evaluated for supporting traditional object detection in decision support scenarios. This study addressed this gap by evaluating three multimodal LLM architectures of different origin specifically for post-detection reasoning and decision support in wildfire reconnaissance.

B. Vision-Language Integration in Object Detection

The integration of vision and language has opened new possibilities in object detection frameworks. For instance, YOLO-World extends the traditional YOLO detector by incorporating vision-language models, allowing it to detect objects described in natural language rather than relying on predefined categories [23]. Similarly, Grounding DINO uses transformer-based architecture with grounded pre-training to detect arbitrary objects based on textual descriptions, showing strong performance without requiring training on specific object classes [24]. These approaches differ from our work when it comes to integration strategy since they embed language understanding directly within the detection model to expand what can be detected. However, the approach in this study keeps detection and language reasoning separate, using the YOLO architecture for spatial detection and multimodal LLMs for post-detection reasoning and explanation. This separation benefits tactical deployment since it allows proven detection models to remain unchanged while adding a modular layer of cognitive support.

C. Explainable AI (XAI) for High-Stakes Decision Support

Research on human-AI collaboration shows that effective decision support requires clear explanations that align with human cognitive processes, in addition to accurate predictions [3], [4], [6]. Senoner et al. demonstrated that visual heatmap explanations greatly improved task performance in manufacturing and medical inspection tasks [3]. Hassan et al. found that explainability in clinical AI encouraged deeper engagement with AI predictions, building trust among users [4]. Recent work by Chen et al. showed that AI confidence levels, text explanations and performance visualizations improved collaborative task performance in healthcare contexts [6]. While XAI benefits are well documented, particularly in healthcare and manufacturing, applications to wildfire detection and decision support systems remains an area not yet explored. Additionally, most existing XAI approaches provide static explanations, such as heatmaps, whereas multimodal LLMs enable dynamic, natural language reasoning based on context. This research explores whether multimodal LLMs can provide more contextually relevant explanations than traditional XAI methods for critical wildfire operations, where operators need actionable recommendations.

D. Wildfire Detection Using Deep Learning

Deep learning has greatly improved automated wildfire detection. Mukhiddinov et al. optimized a YOLOv5-based model for UAV-based smoke detection, achieving high accuracy through model improvements and transfer learning [5]. Similarly, Saydirasulovich et al. improved YOLOv8 with attention mechanisms to create multi-scale smoke detection, reporting an average precision of 79.4% [25]. Beyond model architecture, recent work has shown the importance of data quality. El-Madafri et al. introduced the Wildfire Dataset, which explicitly includes confounding visual elements such as fog, clouds, and challenging lighting conditions that commonly lead to false positives, leading to more reliable detection [26]. While these approaches achieve strong detection performance, they mainly output technical data such as bounding boxes and confidence scores. In real wildfire response operations, such outputs must be interpreted by human decision makers under time pressure, which can increase cognitive load and delay a response. As a result, current wildfire detection systems offer very limited support beyond detection itself, and it remains unexplored how multimodal LLMs can assist in converting detection outputs into natural language explanations for tactical decisions. Our system addresses this limitation by augmenting YOLOv8-based detection outputs with multimodal LLM-generated situation briefs and recommendations, bridging the gap between raw detection data and operator decision-making needs.

E. Edge AI and Model Efficiency

Using multimodal LLMs in wildfire reconnaissance creates major challenges due to hardware limitations, especially when targeting edge devices with limited compute resources. Yao et al. demonstrated that compact multimodal models, such as MiniCPM-V, can achieve performance comparable to larger vision LLMs while running efficiently on mobile devices through model design improvements and compression methods [22]. Building on this, recent surveys have identified key efficiency strategies for multimodal models, including model compression, efficient attention mechanisms and optimized inference pipelines [19]. While these works show that multimodal LLMs can be deployed on edge hardware, the trade-offs between model size, inference speed, accuracy and the quality of generated explanations needs more research. This is relevant for crisis scenarios that require offline operation and quick decision making. Existing studies also mostly evaluate individual models in isolation, leaving open questions about how different multimodal LLMs compare when deployed under similar operational constraints [20], [21], [22]. Our work addressed these deployment constraints by implementing and evaluating all three architectures on identical consumer-grade hardware representative of military field equipment, ensuring findings are directly applicable to resource-constrained environments.

F. Summary & Research Gap

Prior research has advanced vision-language models, edge-deployable AI, and wildfire detection independently, but their

integration for critical decision support remains unexplored. Specifically, no work has systematically evaluated how multimodal LLMs can augment object detection outputs with natural language reasoning under resource-constrained deployment. This thesis addresses this gap by comparatively evaluating three MLLM architectures on identical consumer-grade hardware, measuring their effectiveness in transforming detection results into actionable situation assessments for wildfire reconnaissance.

IV. RESEARCH METHODOLOGY

A. Design Science Research Approach

Following the DSR principles defined by Peffers et al. [27], this study produced a functional artifact in the form of a multi-stage decision-support pipeline, whose value was determined by how well it met stakeholder needs under realistic deployment constraints, rather than by testing predefined hypotheses. Phases within each cycle overlapped rather than proceeded strictly sequentially, with implementation and evaluation informing one another as new findings emerged.

The study was conducted in collaboration with the Swedish Defence Materiel Administration (FMV), which is interested in exploring how multimodal LLMs can augment traditional object detection systems to support decision-making in high-stakes, time-critical scenarios. Wildfire detection served as the concrete use case, but the research questions were framed at the level of architectural viability and trade-offs, making the findings relevant beyond this specific domain.

B. DSR Cycle 1: Requirements and Baseline Evaluation

Build. Requirements were elicited through semi-structured interviews with fire and emergency response professionals specializing in wildfires ($n=4$), identifying operational decision-making priorities and contextual information needs. Based on these requirements, a three-stage MVP pipeline was implemented combining image quality screening, YOLO-based object detection, and multimodal LLM reasoning, with three LLM architectures selected to span different geographic origins and technical approaches.

Evaluate. The pipeline was evaluated through quantitative benchmarking across the three LLM architectures on identical consumer-grade hardware, measuring classification accuracy, inference speed, and memory usage. Qualitative evaluation was not conducted in Cycle 1. The prototype served as a working baseline and measurable starting point, with stakeholder artifact evaluation deferred to Cycle 2 where the system is more complete.

C. DSR Cycle 2: Refinement and Evaluation

Reflect. Cycle 1 confirmed that all three LLM architectures could perform binary fire classification on consumer-grade hardware within acceptable memory constraints, and that 3–4B parameter variants performed comparably to larger ones for this task. Stakeholder interviews conducted in parallel surfaced concrete operational priorities such as water supply, road accessibility, terrain, settlements, and wind that could not

be incorporated before the Cycle 1 prototype was delivered due to scheduling, making Cycle 2 the natural point of integration.

Build. The artifact was extended with a geographic context pipeline querying pre-loaded OSM and terrain data, a refined system prompt supporting structured tactical output, and an operator-facing user interface. Requirements were revised to reflect stakeholder findings from Cycle 1.

Evaluate. The completed artifact was evaluated through two complementary means: a full quantitative benchmark across all three LLM architectures under both YOLO and no-YOLO conditions, and qualitative evaluation interviews with fire and emergency response professionals assessing the artifact’s usefulness, interpretability, and operational fit.

D. Data Collection

Data collection takes two forms: quantitative benchmark data and qualitative interview data. Cycle 1 interviews served requirements elicitation and Cycle 2 interviews evaluated the completed artifact.

1) *Cycle 1 Benchmark:* Two separate benchmarks were conducted in Cycle 1. The YOLO object detection stage was evaluated independently as an image-level wildfire detector on a balanced sample of 100 images from the El-Madafri wildfire dataset, consisting of 50 fire and 50 non-fire images (see Appendix I) [26]. An image was counted as a positive detection if YOLO identified at least one Fire or Smoke object.

The three MLLM architectures were benchmarked separately against the same dataset on an Apple Silicon MacBook Air M2 with 16GB RAM, without YOLO preprocessing to isolate baseline vision-language reasoning. The accuracy benchmark sampled 100 images per model, and the performance benchmark sampled 15 images with cooldown periods between inferences to control for thermal throttling. A fixed random seed and shared system prompt were used across all models to ensure reproducibility. Captured metrics for both benchmarks include accuracy, precision, recall, F1, and Matthews correlation coefficient. The YOLO benchmark additionally captures frames per second, and the LLM benchmark captures inference duration, tokens per second, and peak runtime memory.

2) *Cycle 2 Benchmark:* The Cycle 2 benchmark extended Cycle 1 in three ways: the sample size was doubled (200 images for accuracy, 30 for performance), each benchmark was run twice, with YOLO preprocessing on/off to allow direct comparison under otherwise identical conditions, and a dedicated scenario evaluation script was added that runs all models against each of the operational scenarios. A single pre-computed scenario context was held constant across all 200 accuracy benchmark images so that geographic context was not a variable between images. All runs were performed on the same hardware class as Cycle 1.

3) *Qualitative Interviews:* Cycle 2 evaluation interviews were conducted in person with the same wildfire response professionals who participated in the Cycle 1 requirements elicitation ($n=4$). Interviews followed a semi-structured protocol and included a live demonstration of the completed artifact.

Participants were asked to assess the system’s usefulness, output interpretability, and operational fit. The interview protocol is provided in Appendix H

E. Data Analysis

1) *Quantitative Benchmark Analysis*: Classification performance metrics (accuracy, precision, recall, F1, and Matthews correlation coefficient) were computed per model from the benchmark runs. For Cycle 2, inference failure counts were filtered out before metric computation and reported separately to avoid distorting performance figures. Timing metrics were summarised using median and interquartile range across performance benchmark runs to remain robust to outliers from background system activity or residual thermal variation. Results are compared across LLM architectures as well as across YOLO and no-YOLO conditions, and visualised as bar charts and confusion matrices.

MCC is inherently designed for binary classification and is highly appropriate given the binary fire or no-fire prediction schema utilized across all models and cycles. It is particularly valuable because local LLM architectures can produce heavily skewed prediction distributions when faced with complex visual classification tasks. In scenarios where a model experiences collapse and default-predicts a single class, standard accuracy figures can be highly deceptive even when utilizing a balanced dataset. Because MCC penalizes both false positives and false negatives symmetrically, it ensures a reliable, single-number summary of true predictive power that successfully exposes whether an architecture has collapsed or maintains genuine discriminative utility.

2) *Qualitative Interview Analysis*: Cycle 1 interview transcripts were analysed through content analysis to extract operational priorities and contextual information needs, informing the Cycle 2 requirements set. Cycle 2 evaluation interview transcripts were similarly analysed to identify themes relating to artifact usefulness, output interpretability, and operational fit. In this context, usefulness is defined as the degree to which the system output supports a responder in making faster or better-informed initial decisions, specifically whether the generated recommendations reference actionable and verifiable information that the operator could not have readily derived from the image alone.

3) *Qualitative Output Analysis*: Decision support output quality was analysed by the research team through direct inspection of LLM outputs produced during evaluation. Each model’s outputs were assessed against the same operational context block, examining whether recommendations referenced named locations, distances, and supply categories from the context, whether situation briefs were concise and radio-ready, and whether key constraints correctly identified operational limitations. This analysis was conducted independently of the stakeholder interviews. Interviewees evaluated output from a single model, while the team’s own analysis compared output quality across all three models.

F. Process

The five-member engineering team worked on a weekly sprint cadence with daily standups, sprint planning on Mondays, and retrospectives on Fridays, collaborating primarily on-site with asynchronous communication via Discord. Development followed a feature-branch model on GitLab with a protected main branch, merge request reviews requiring at least one reviewer, and tasks tracked as issues on a Kanban board using shared templates with user stories and acceptance criteria tied to one of two DSR cycle milestones. The thesis manuscript was written collaboratively in Overleaf alongside artifact development, with shared working documents in Google Drive, allowing implementation insights to inform the manuscript concurrently. Stakeholder interview participation varied across team members throughout the project.

V. THE ARTIFACT

A. System Architecture Overview

The software artifact is structured as a multi-stage pipeline, which was extended and improved over the two research cycles. A high-level minimum viable pipeline architecture from Cycle 1 is shown in Fig. 1, and the finalized Cycle 2 architecture is shown in Fig. 2.

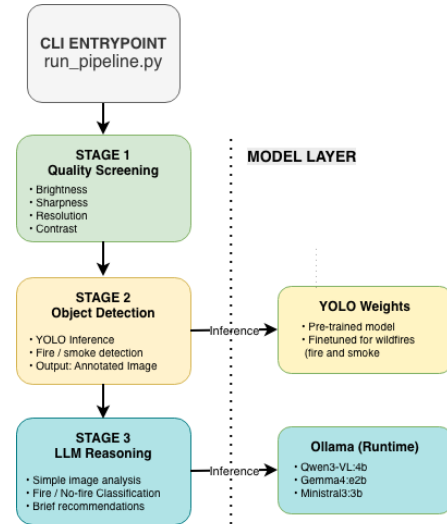


Fig. 1: System architecture for Cycle 1

B. Hardware Environment

All components of the artifact are developed and benchmarked on Apple Silicon MacBooks (M2 generation), reflecting the hardware available to the research team. This hardware class represents a common profile of capable consumer-grade laptops: unified memory (16 GB), passive cooling on some models, and general-purpose CPU and GPU compute without specialized accelerators.

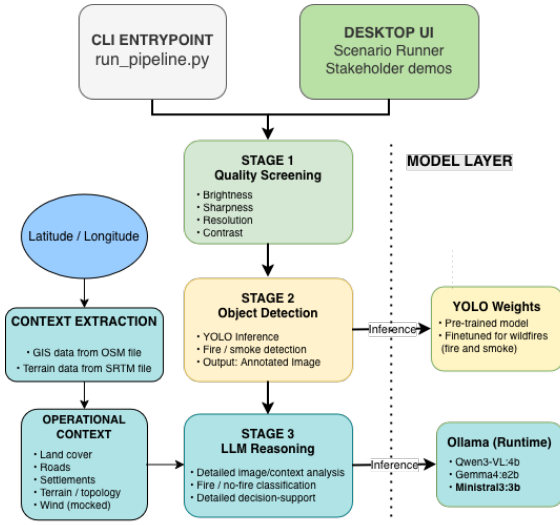


Fig. 2: System architecture for Cycle 2

C. Cycle 1: Minimum Viable Pipeline

Cycle 1 establishes a three-stage pipeline for ingesting aerial wildfire imagery, detecting fire and smoke, and producing a natural-language classification and recommendation via a multimodal LLM. The primary objective is to demonstrate feasibility on consumer-grade edge hardware and establish a benchmarking framework for comparing model architectures. The full requirement list and traceability are provided in Appendix A.

1) *Stage 1: Image Quality Screening*: The first stage implements **REQ01** by filtering out images whose characteristics are likely to make downstream detection and LLM reasoning unreliable. It is implemented using OpenCV and applies four checks in sequence: resolution, brightness, contrast, and sharpness.

Resolution is checked first as it is computationally inexpensive. Brightness is computed as mean pixel intensity but allows images with low average brightness to pass if they contain sufficient localised bright content. This preserves images with visible fire regions that would otherwise be rejected on overall brightness alone, including the nighttime fire images present in the dataset. Contrast is measured as the standard deviation of pixel intensity and bounded at both ends.

Sharpness was initially implemented using the variance of the Laplacian, a common blur detection measure. This proved unreliable for wildfire imagery, where smoke and haze produce naturally soft regions that are semantically useful but receive low sharpness scores. The check was redesigned as a tile-based Tenengrad method with texture filtering: the image is divided into tiles, low-texture tiles (smoke, sky, haze) are excluded from evaluation, and sharpness is aggregated only over tiles with sufficient texture content [28].

All threshold values are configurable. If any check fails, the image is rejected and a diagnostic record is returned indicating which checks failed, allowing the caller to report specific rejection reasons rather than a binary outcome.

2) *Stage 2: Object Detection*: The second stage of the pipeline implements **REQ02** by detecting fire and smoke from imagery that has passed quality screening. The stage uses YOLOv8, a widely-adopted single-stage object detection architecture [29]. The specific model is an open-weights variant fine-tuned for forest fire and smoke detection [11], accessed at commit `a44e1ecdb84b06bda145b1c3ad53854046fe14dd` and run with default YOLOv8 inference parameters without modification. The model produces bounding boxes for two classes: *Fire* and *Smoke*. The SHA-256 checksum of the weight file used is provided in Appendix J for reproducibility.

Selecting an appropriate detection model required several iterations. YOLO-MP [30] was initially considered due to its lightweight design targeting forest fire detection, but the accompanying repository did not include the custom modules required to instantiate the model, making the published weights unusable in practice. A general-purpose YOLOv8 model was subsequently tested but showed poor accuracy in distinguishing fire and smoke from visually similar phenomena common in the wildfire dataset. The final choice previously mentioned, a pretrained YOLOv8 weights fine-tuned specifically for fire and smoke detection, produced reliable detections on the evaluation imagery and was adopted as the default.

The detection stage receives an image as input, runs inference, and produces an annotated version of the image with labeled bounding boxes drawn over any detected regions of fire or smoke. Each label visually displays the class and its confidence score. This annotated image is written to a known output path and serves as the handoff to Stage 3. In Cycle 1, no structured detection metadata is passed to the LLM reasoning stage. Confidence scores and other detection metadata are therefore available to the LLM only as visual elements of the image when YOLO preprocessing is enabled, rather than as structured data it can reason over directly. This is revisited in Cycle 2 (Section V-D).

3) *Stage 3: Multimodal LLM Reasoning*: The third and final stage of the pipeline produces a natural-language classification, reasoning, and recommendation based on the annotated image received from Stage 2. This stage is responsible for satisfying **REQ13** through the reasoning field, and for producing the classification and recommendation fields that the pipeline is designed to deliver. In this stage, the three LLM families are supported and compared: Ministral 3, Qwen3-VL, and Gemma 4. The specifics of model selection and their characteristics are described in Section V-C4.

Models are served locally via Ollama version 0.20.3 [31], a local LLM runtime that exposes a uniform HTTP API for loading and querying models from different families. Serving models locally removes the need for network connectivity at inference time, satisfying **REQ11**. Two alternatives were considered: PyTorch and MLX. Ollama was selected over PyTorch + MPS and MLX because it provides a uniform serving layer across all three families with mature multimodal support which aligns with the modular-integration requirement (**REQ09**). MLX also lacked image support. The system prompt is stored in a separate file and loaded at runtime, allowing prompt

iteration without changes to the pipeline code.

The stage uses Ollama’s structured output capability to enforce that the LLM returns valid JSON matching a predefined schema. The schema includes three fields: a classification (one of *fire_detected*, *no_fire_detected*, or *uncertain*), the reasoning supporting the classification, and a tactical recommendation for the operator. This design guarantees that downstream consumers receive parseable structured output rather than relying on prompt discipline alone, and removes the need for parsing of free-form text. The *uncertain* classification is included for situations where visible evidence does not support a confident binary decision.

The LLM is invoked with the temperature parameter set to zero, which produces deterministic output for a given input and supports the reproducibility requirement defined in **REQ08**. The response from the LLM is validated against the expected schema before being returned and any deviation from the schema raises an error rather than being silently accepted.

4) *Model Selection*: Three multimodal LLMs were selected for comparative evaluation: Ministral 3 (Mistral, France/EU), Qwen3-VL (Alibaba, China), and Gemma 4 (Google, United States). Each is introduced in Section II-C. Model integrity can be verified using the Ollama digest hashes provided in the appendix (Appendix J). The rationale for choosing these three is described below.

The selection was driven by three considerations. First, the research scope targets edge-deployable models, which restricts the candidate set to open-weight architectures that can run locally on consumer-grade hardware. Second, the research aims to compare architectures of differing origin to avoid overrepresenting any single research lineage. The three selected families originate from three distinct organizations across three regions. Third, the evaluated variants should be parameter-comparable so that observed differences reflect architectural rather than scale differences.

A to fix preliminary test was performed on a hand-selected subset of 40 images (20 fire and 20 non-fire) across all six variants offered by the three families, covering both the smaller (2–4B) and larger (7–8B) variants of each. The test indicated that for binary fire classification with short natural-language reasoning, the only evaluation dimension available at that point, the larger variants did not yield improvements in classification accuracy or reasoning output proportional to their higher inference cost. The smaller variant from each family was therefore selected for the full evaluation. The impact of model size on decision support output quality, which only became relevant once geographic context and a richer prompt were introduced in Cycle 2, was not assessed at this stage.

Achieving comparability across the three selected variants required some care. Ministral 3 and Qwen3-VL are both dense transformer architectures and their parameter counts (3B and 4B respectively) reflect both the stored and the activated parameters during inference. Gemma 4 E2B, by contrast, uses a Per-Layer Embedding (PLE) design: the variant carries 5.1B stored parameters but limits active computation to roughly

TABLE I: Selected models for comparative evaluation.

Family	Origin	Params	Size	Architecture
Ministral 3	EU	3B	3.0 GB	Dense
Qwen3-VL	China	4B	3.3 GB	Dense
Gemma 4 E2B	US	2.3B*	7.2 GB	PLE

*Effective; 5.1B stored. Full SHA-256 digests in Appendix J.

2.3B effective parameters at inference time [16]. This places Gemma 4 E2B within the same computational footprint as the 3B and 4B dense models used for comparison, despite its larger disk size. Reporting all three models together is therefore meaningful on inference-cost grounds, even though the raw parameter counts appear divergent.

An earlier candidate for the United States representative was LLaVA [20], which has historically been one of the most widely cited open-weight multimodal models. LLaVA was ultimately excluded because its last substantive release predates the other three families by more than a year. Gemma 4, released during the research period, was selected in its place as a more current representative of United States open-weight multimodal development for edge deployment. This substitution keeps the comparison focused on contemporary architectures rather than mixing recent releases with a model whose development has largely stalled.

5) *Benchmarking Framework*: A dedicated benchmarking framework was developed alongside the pipeline to evaluate the three selected LLMs under controlled and reproducible conditions, implementing the benchmarking requirements **REQ04–REQ10**. The framework operates outside the live pipeline: it drives LLM inference directly using a shared system prompt, fixed random seed, and identical sampled image set across models, producing per-image metrics recorded to disk for later analysis.

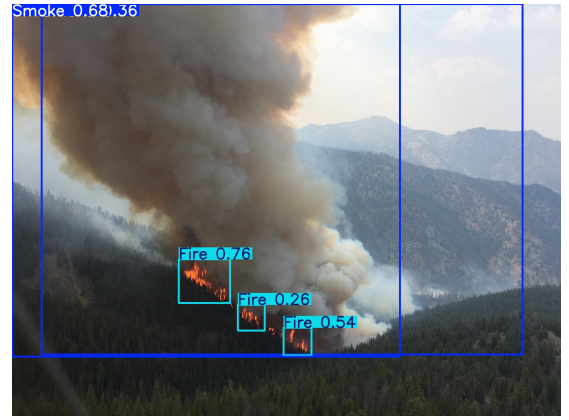
The framework supports two complementary modes. Accuracy mode runs each model over a large sample of the dataset without cooldowns, optimizing for statistically meaningful classification metrics. Performance mode runs each model over a smaller sample with configurable cooldown periods between images and models, ensuring each inference starts from a comparable thermal state and producing the timing measurements used for inference-speed comparisons.

Both modes draw from the wildfire dataset [26] using balanced random sampling with a fixed random seed. An optional YOLO preprocessing stage can be enabled, passing each image through object detection before the LLM, with YOLO timing and detection count recorded separately. An analysis script reads all benchmark CSVs and produces classification performance metrics, response characteristics, and timing breakdowns forming the raw material for the quantitative results in Section VI.

6) *Implementation Details*: The artifact is implemented in Python, with each pipeline stage developed as a separate module under a shared *pipeline/* package to reinforce separation of concerns and enable modular substitution (**REQ09**). The benchmarking framework lives in a separate top-level



(a) Input image from the `/train/fire` directory of the Wildfire Dataset (37325108026_bb99689e1f_o.jpg).



(b) YOLO-annotated image with fire and smoke bounding boxes.

Fig. 3: Example image input and output: (a) input image from the wildfire dataset [26], (b) YOLO-annotated image with fire and smoke bounding boxes.

```
{
  "classification": "fire_detected",
  "reasoning": "Visible flames and dense smoke plumes, including active fire fronts with high confidence annotations (0.68, 0.76, 0.26, 0.54), indicate active wildfire presence.",
  "recommendation": "Immediately escalate to incident command. Deploy suppression resources (firefighters, air assets) and coordinate with local authorities to contain the fire perimeter. Prioritize evacuation of nearby communities and monitor smoke drift patterns."
}
```

Fig. 4: Example LLM output from the Ministral3:3b model during Cycle 1 for the image in Figure 3.

Full model SHA-256 digests are provided in Appendix J

directory and imports pipeline modules as needed, ensuring that benchmarking concerns do not leak into production code paths. Dependency management is handled through *uv*, which produces lockfile-based reproducible environments across machines.

Each stage exposes both a command-line entry point for isolated testing and a programmatic interface for use within the orchestration layer. The orchestration module composes the three stages end-to-end, taking an image path as input and passing the output of each stage to the next. This separation allows individual stages to be exercised independently of the full pipeline.

7) *Cycle 1 Outcomes and Identified Limitations*: Cycle 1 delivered a working minimum viable pipeline satisfying the functional, benchmarking, and non-functional requirements defined at the start of the cycle, with the benchmarking framework producing reproducible quantitative results across all three LLM architectures.

Alongside these outcomes, three limitations shaped the direction of Cycle 2. First, the LLM reasoning stage receives only the annotated image and has no access to structured detection metadata or geographic context, keeping output close to describing what is visible rather than offering grounded

tactical decision support. Second, stakeholder interviews overlapped with implementation work, and their findings could not be incorporated before the minimum viable pipeline was delivered. Third, qualitative evaluation of decision support quality was not conducted in Cycle 1, as interviews were allocated to requirements elicitation. All three limitations were addressed in Cycle 2.

D. Cycle 2: Refinement and Extended Reasoning

Cycle 2 extends the pipeline toward operational decision support, grounding LLM recommendations in geographic and environmental context rather than image content alone. Requirements were shaped by stakeholder interviews from Cycle 1, which surfaced concrete operational needs: geographic context integration, structured output, offline operation, and a map view. The full requirement list and traceability are provided in Appendix A and their related user stories for both cycles are listed in Appendix B.

1) *Quality Screening Refinement*: Early testing showed the initial thresholds were too strict. The brightness check was revised to use a localised criterion alongside overall brightness, allowing dark images with visible fire regions to pass. Contrast thresholds were adjusted through manual testing and the

minimum resolution was lowered to 300×300 pixels, reducing unnecessary rejections while preserving filtering of genuinely unusable imagery.

2) *Geographic Context Pipeline*: The central contribution of Cycle 2 is a structured geographic context pipeline that satisfies **REQ14** by querying pre-loaded local data for a given coordinate pair and producing an *OperationalContext* object passed to the LLM alongside the fire image, enabling recommendations that reference specific locations and distances rather than generic guidance.

Data Sources: Two offline data sources feed the pipeline. OpenStreetMap (OSM) vector data, downloaded as a regional *.osm.pbf* extract from Geofabrik, provides roads, water bodies, settlements, buildings, power lines, and protected areas. Elevation data is sourced from the Shuttle Radar Topography Mission (SRTM) at 30m resolution, downloaded via the *elevation* Python package. Lantmäteriet’s GSD-Höjddata was the original target for terrain data, as it offers significantly higher resolution over Sweden, but obtaining and integrating the data proved impractical within the project timeline due to the complexity of their access process. SRTM was adopted as a practical alternative. At 30m resolution it is sufficient to characterise slope steepness and aspect at the operational scale relevant to fire response.

All data is pre-loaded onto the device and queried offline at runtime, satisfying **REQ19**. No network access is required after the initial data download.

Schema Design: The extracted data is validated against a Pydantic schema (*OperationalContext*) that defines the contract between extraction, formatting, and LLM inference. The schema captures: coordinates and administrative region, land cover type, terrain (elevation, slope in degrees, slope steepness category, and aspect), water sources with supply category and road accessibility distance, primary road access and nearby tracks with vehicle accessibility flags, settlements with type and distance, assets at risk (building count, permanent structure flag, power line presence, and protected area name), named natural features, and optional wind data.

Water sources are classified as *heavy supply* (lakes above 10,000 m², reservoirs, rivers) or *light supply* (small ponds, streams, ditches, wetlands). This distinction was identified by fire professionals as operationally critical during interviews. Road accessibility is derived conservatively from OSM highway, surface, and track type tags: tracks without a confirmed solid surface or appropriate track grade are treated as not confirmed accessible. When no data is available, the conservative assumption is that the route is impassable.

Fire stations within 30 km were added to the schema following an observation that responders benefit from knowing the nearest staging resources. The formatter includes station names and distances in the context block when present. The extraction also computes a fire behavior synthesis note combining wind direction, terrain aspect, slope steepness, and land cover fuel type, for example, flagging when wind drives fire upslope on a steep SW-facing slope, or when wetland fuel reduces expected fire intensity. This note is appended to the context block to

make it easier for the small models to combine differing contextual data into usable decision-support regarding fire spread.

Extraction: The extraction script loads the *.osm.pbf* once per process into in-memory spatial data structures. All calculations are performed in a projected coordinate system appropriate for Sweden. Administrative region lookup uses convex hull approximations of municipality boundaries, a deliberate approximation that works reliably at municipality scale but occasionally misses county edges. Land cover priority rules prevent small wetland polygons from overriding larger surrounding forest areas, and water sources are ranked by a combined score weighting distance and road accessibility.

Terrain: Terrain properties are computed from a small window centred on the observation coordinate using finite differences on the SRTM raster. Elevation is read directly from the centre pixel. Slope is computed from the gradient magnitude and categorised into five operational levels: flat, gentle, moderate, steep, and very steep. Aspect is derived from the gradient direction and expressed as an eight-point compass bearing.

Context Formatting: The *OperationalContext* object is serialised to a plain-text block by a dedicated formatter before being passed to the LLM. The formatter produces a consistently structured output of approximately 150–220 tokens covering all populated fields. Extraction metadata is excluded from the formatted block and is instead retained in the on-disk JSON for reproducibility but serves no purpose in the LLM prompt.

Wind direction is formatted to include both the source bearing and the computed spread direction, for reasons discussed in Section V-D4.

3) *Scenario Design*: To evaluate whether the LLM produces meaningfully different recommendations given different geographic contexts, six scenarios were designed covering distinct operational situations. Each scenario consists of a pre-computed *OperationalContext* JSON and a paired wildfire image, stored together in a dedicated folder. The image, sourced from the existing wildfire dataset and supplementary internet searches, does not need to exactly match the geographic location but is selected to be representative of the scenario type.

Scenarios were chosen to exercise different combinations of context features, ensuring no two scenarios produce similar context blocks. Coordinates were identified by inspecting OpenStreetMap and Google Maps and verified by running the extraction pipeline and reviewing the resulting context JSON.

Wind is assigned once per scenario as a mocked value and persists across all evaluations using that scenario, ensuring consistent conditions. This allows wind to contribute to the LLM’s reasoning without introducing variability between runs.

The six scenarios and their defining characteristics are summarized in Table II.

4) *LLM and Prompt Engineering*: The system prompt and LLM inference configuration evolved substantially across Cycle 2, driven by observed model behavior. Prompt versions are

TABLE II: Scenario overview

Sc.	Loc.	What it tests
01	Acktjärnsåsarna, Västmanland	Water availability vs difficult access; protected area
02	Tyresta NP edge, Stockholm	Wildland-urban interface; 83 buildings within 2 km
03	Skuleskogen, High Coast	Steep terrain; terrain-driven fire behaviour and access difficulty
04	Skåne countryside	Dense assets with good road access but no heavy water supply
05	Vakö myr, Kronoberg	Smouldering peat risk; protected mire with light supply water only
06	Sorsele, Västerbotten	Remote forest; sparse infrastructure and minimal OSM coverage

preserved as numbered files for reproducibility, and benchmark runs reference the specific version used. The system role content was separated from the task prompt into a dedicated file, allowing the role definition to be adjusted independently. All models were queried with temperature set to zero.

Cycle 1 Baseline: The Cycle 1 prompt was minimal: classify the image, return JSON with classification, reasoning, and recommendation. It allowed an *uncertain* classification-alue, which was dropped in Cycle 2 to force a binary commit. Ambiguous cases are expressed in the reasoning field rather than encoded in the classification.

Introducing geographic context: The most significant prompt change in Cycle 2 was framing how the model should reason over two complementary inputs. A deliberate design decision was made not to correlate the image with specific context features. This was motivated by two factors: the image captures a much smaller spatial area than the context describes, and sourcing wildfire imagery for specific geographic locations is not feasible given the time frame. The system is therefore designed to treat image and context as describing the same operational situation without requiring visual correspondence. An explicit instruction was added to this effect, framing the inputs as: the image tells you what is happening, the context tells you where and what surrounds it.

Source attribution was added to make the evidence chain explicit. Models prefix image observations with “From the image...” and context observations with “From the context...”, which also surfaces cases where the model attributes fabricated claims to the context.

Context bias and image primacy: Introducing operational context caused models to classify regular forest images as fire detected even when no fire was visible. To try and mitigate this, an image-context separation clause was added: the classification must be based solely on visual evidence, and the context must not influence the fire or no fire determination.

Output structure changes: A *tactical_priority* field was introduced to provide structured guidance, and the recommendation field was changed from free-form text to action-oriented bullet points for improved scannability and actionability. In practice the priority field proved unreliable: models defaulted to suppress regardless of context, and adding explicit decision criteria only partially improved this. More significantly, the

field was sensitive to changes that had nothing to do with the prompt content itself since moving the system prompt between roles changed the tactical priority output without any instruction change. This volatility made the field unsuitable as a reliable signal, and it was removed from the schema, leaving **REQ15** descoped. Prioritisation reasoning remains present in the free-text recommendation field, which was retained in bullet point format. A *key_constraints* field was subsequently added to surface specific operational limitations in structured form, which consists of up to three short strings naming concrete constraints on response operations such as single road access, a settlement downwind, or a power line helicopter constraint. Fire behavior observations are explicitly excluded from this field and belong in the reasoning field instead. The structured output schema: *classification, reasoning, recommendation, situation brief, and key constraints*, satisfies **REQ16**.

Wind direction: Models consistently misinterpreted meteorological wind direction as spread direction. A south-westerly wind blows *from* the south-west, meaning fire spreads toward the north-east, but models treated the source bearing as the direction of spread. Correcting this through prompt instructions alone proved unreliable. The fix was applied at the formatter level, where the spread direction is computed explicitly and included alongside the source bearing.

System role and modularisation: As the task prompt grew richer across iterations, Qwen3-VL 4B began producing empty or malformed responses on certain image and prompt combinations. Investigation pointed to the system role as a contributing factor. The model’s thinking mode appeared to be triggered more aggressively by instructions in the system role than equivalent content in the user role. Moving the full task prompt to the user message, with only a concise role definition retained in the system role, was adopted as the default configuration after the change produced no observable effect on Ministral or Gemma output quality. The system role content was extracted into a separate versioned file to allow independent adjustment without versioning the full task prompt. The change yielded a slight improvement in output throughput for Qwen3-VL, but reliability issues remained, meaning **REQ17** is only partially met, as reported in Section VI-B3.

5) Benchmarking Extensions: The Cycle 1 benchmarking framework was extended with GIS context support, a dedicated scenario evaluation script running all models against all six scenarios, and updated CSV output capturing the new structured fields. Inference failures are logged and reported separately to avoid distorting accuracy figures.

A dedicated scenario evaluation script was added alongside the main benchmark. Rather than running many images against one scenario, it runs all models against all six scenarios, one image per scenario, and records the full structured output including reasoning, recommendation, and situation brief. This supports qualitative evaluation of how well each model utilizes geographic context across distinct operational situations.

CSV output and the analysis script were extended to capture the new fields and failure counts, ensuring inference failures

are logged and reported separately rather than silently distorting accuracy figures.

6) *User Interface*: Cycle 2 extended the artifact with an operator-facing user interface implemented in PySide6, running locally on the same edge device as the pipeline. Its purpose is to make pipeline outputs inspectable during stakeholder evaluation and to visualise what a potential operational system could look like.

The interface combines a scenario map with the pipeline output panel. The map satisfies **REQ20** by displaying available scenarios as points of interest on an overview map of Sweden, switching to a local OSM-rendered map with pan and zoom when a scenario is loaded. The pipeline panel presents quality screening results, the YOLO-annotated image, and the full LLM output including classification, situation brief, key constraints, reasoning, and recommendation.

The interface supports two modes of use. In scenario mode, selecting a scenario automatically loads the paired image and pre-computed context, providing a consistent setup for stakeholder evaluation. In testing mode, the user can freely select images, toggle YOLO and wind, switch prompt versions, and adjust context, allowing experimental configurations to be evaluated without changing pipeline code. LLM inference runs in a background thread to keep the interface responsive, with cancellation support that preserves completed upstream stage results. Default settings satisfy **REQ18** by requiring no mandatory input beyond selecting a scenario.

VI. RESULTS

A. Baseline Findings

1) *Object Detection Performance*: The detector achieved 86.0% accuracy, 81.0% precision, 94.0% recall, and an F1 of 87.0%. As shown in Fig. 8b, YOLO correctly detected fire in 47 of 50 fire images and correctly rejected 39 of 50 non-fire images, producing 11 false positives. Fig. 8a shows that fire images generated substantially more detections per image than non-fire images, consistent with the lower false positive rate on fire images. The median processing time was 52.94 ms per image (18.89 FPS), with inference accounting for the largest proportion of runtime as shown in Fig. 8c.

Fire images in the benchmark varied substantially in fire extent. The coverage distribution is discussed as a threat to validity in Section VII-C8.

2) *LLM Classification Performance*: The share of predictions in each outcome category per model is illustrated in Fig. 9a. Qwen3-VL 4B achieved 99% raw accuracy, Ministral 3B 90%, and Gemma 4 E2B 82%. Gemma was the only model to produce ambiguous predictions, abstaining on 13% of images. Ministral and Qwen3-VL committed to a binary classification in every case.

Fig. 5a reports precision, recall, and F1 computed on confident predictions only. Qwen3-VL 4B achieves the highest scores across all metrics, followed by Gemma 4 E2B. Raw counts of true/false positives and negatives are shown in Fig. 5b. Gemma’s 13 ambiguous cases and Ministral’s 7 false positives account for the majority of the difference between

the models’ confident-prediction accuracy. Finally, Fig. 9b reports the Matthews Correlation Coefficient where Qwen3-VL achieves near-perfect correlation at 0.98, Gemma 0.89, and Ministral 0.80.

3) *Inference Speed and Resource Use*: Median total inference time per image is shown in Fig. 5c. Gemma 4 E2B is the fastest at 14.7 seconds per image, followed by Ministral 3B at 19.0 seconds, with Qwen3-VL 4B the slowest at 25.5 seconds.

Generation speed in tokens per second is reported in Fig. 9c. Gemma 4 E2B and Ministral 3B generate at comparable rates (~ 34 tokens/s), while Qwen3-VL is slower at 26.1 tokens/s. Fig. 5c shows median total inference time and its breakdown into prompt/image processing, text generation, and Ollama overhead. Resource utilization is compared in Fig. 5d, showing that all models remain well below the 16 GB RAM threshold. Finally, Fig. 9d shows the distribution of response lengths. Ministral produces the longest responses (median 62 words).

B. Refinement Findings

1) *Classification Accuracy*: The results show substantial differences between the three evaluated architectures. Qwen3-VL 4B achieved the strongest classification performance on valid outputs, with 95.8% accuracy without YOLO and 93.2% with YOLO-annotated images, along with the highest F1 and MCC scores in both configurations (Figs. 10a, 6a, 10b). However, the model failed to produce parseable output in 56 of 200 cases without YOLO and 53 of 200 cases with YOLO. These failures were not concentrated in one ground-truth class. Without YOLO, Qwen3-VL failed on 25 of 100 fire images and 31 of 100 non-fire images and with YOLO, it failed on 25 of 100 fire images and 28 of 100 non-fire images (Table IV). Garbled outputs that still contained a valid classifications were not counted as failures. When parseability failures are counted as incorrect predictions, Qwen3-VL’s effective full-run accuracy falls to 69.0% without YOLO and 68.5% with YOLO. (Figs. 6b, 6c).

Gemma 4 E2B was substantially more reliable, producing valid output on all 200 images in both configurations. Without YOLO it achieved 85.0% accuracy, 100.0% precision, 70.0% recall, an F1 of 82.4%, and an MCC of 0.734 (Figs. 10a, 6a, 10b). With YOLO annotations, accuracy fell to 77.5% as recall dropped to 55.0% (Fig. 6c), meaning Gemma produced no false positives but missed a notable share of true fire images.

Ministral 3B showed the weakest classification performance, achieving 53.0% accuracy without YOLO and 53.5% with YOLO (Fig. 10a). Despite near-perfect recall of 99.0% and 100.0% respectively, this came at the cost of 93 false positives in both configurations (Figs. 6b, 6c), making the model strongly biased towards classifying images as fire.

The effect that YOLO annotations produced was mixed and did not produce a general improvement in classification accuracy. For Ministral, YOLO annotations slightly increased accuracy from 53.0% to 53.5%, but the improvement was negligible and did not address the model’s false-positive bias. For Gemma, YOLO annotations reduced accuracy from 85.0%

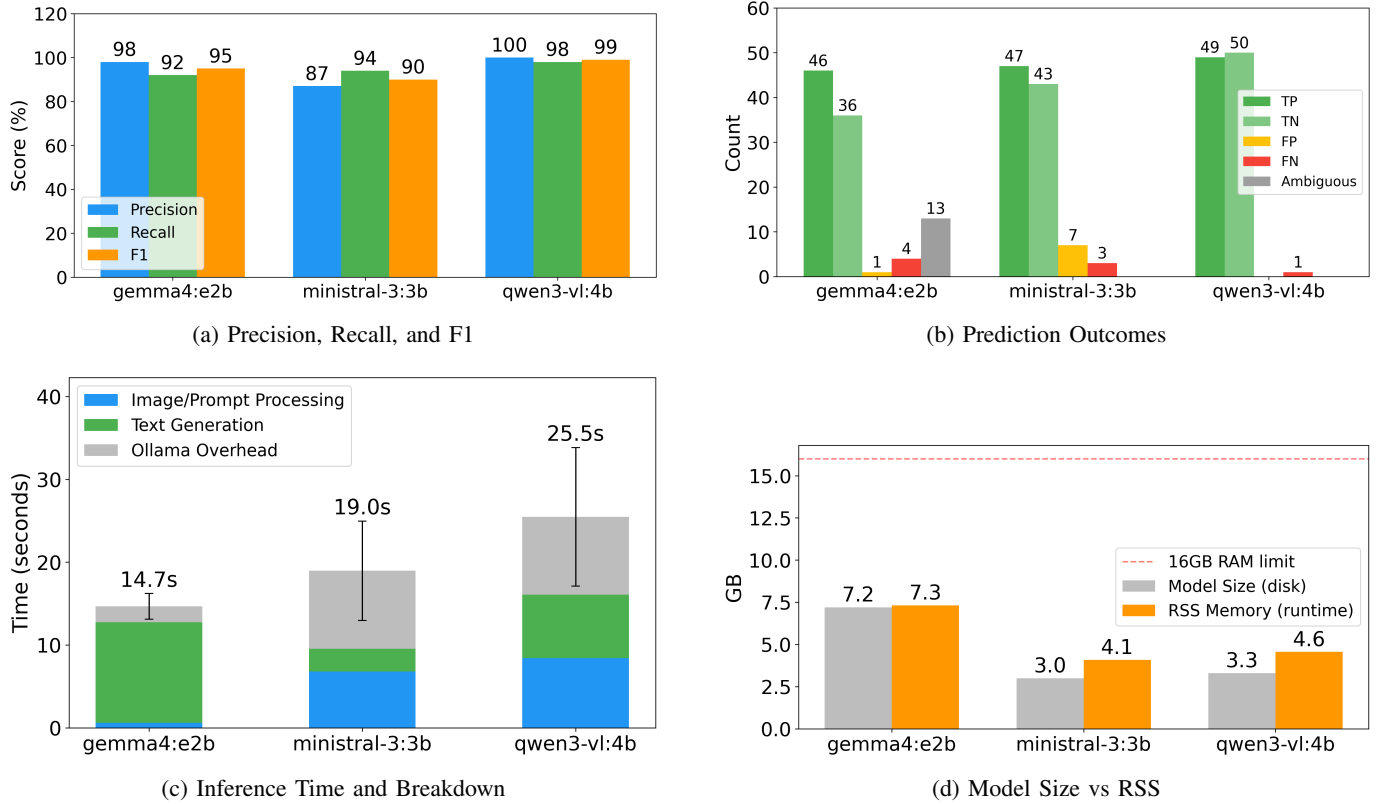


Figure 5: Cycle 1 LLM benchmark results — classification performance and resource utilisation.

to 77.5%. For Qwen3-VL, valid-output accuracy decreased from 95.8% to 93.2%, while the failure rate remained high. These results suggest that visually annotating the image with YOLO detections does not automatically improve multimodal LLM classification, and may in some cases interfere with the model’s own visual interpretation.

2) *Inference Speed and Resource Use:* Median total inference time per image is shown in Fig. 6d. Gemma 4 E2B and Ministral 3B showed comparable latency at 30.9s and 32.3s respectively without YOLO, and 32.9s and 32.0s with YOLO-annotated images. Qwen3-VL 4B was substantially slower in both configurations at 118.5s without YOLO and 113.5s with YOLO. Across all three models, the addition of YOLO preprocessing produced no meaningful change in total inference time. Generation speed in tokens per second, reported in Fig. 10c, was similarly stable across both configurations, with Ministral at 32.1 tokens/s, Gemma at 29.9 tokens/s, and Qwen3-VL at 24.2–24.9 tokens/s. Fig. 6d shows that for Gemma and Ministral, the image/prompt processing component is modestly larger in the YOLO configuration than without. For Qwen3-VL, Ollama overhead accounts for the large majority of total inference time in both configurations.

The Cycle 2 memory results should be interpreted with caution for Gemma 4 E2B. Its reported RSS values were 1.1 GB without YOLO annotations and 1.3 GB with YOLO annotations, which is anomalously low relative to both its 7.2 GB model size and its Cycle 1 memory measurement. The

cycle 2 Gemma memory measurement is therefore not treated as reliable evidence of runtime memory usage.

3) *Inference Reliability:* While Ministral-3 3B and Gemma 4 E2B produced valid structured output on all images in both configurations, Qwen3-VL 4B exhibited substantial inference failures. Table III shows the breakdown across both runs.

TABLE III: Qwen3-VL 4B inference outcome breakdown across accuracy benchmark runs.

Run	Timeout (240+ s)	Empty	Invalid structure	Garbled content	Valid output
No YOLO	1.5%	20.0%	6.5%	9.0%	63.0%
YOLO	5.0%	16.0%	5.5%	12.0%	61.5%
Combined	3.25%	18.0%	6.0%	10.5%	62.25%

TABLE IV: Qwen3-VL 4B hard failure rate by ground-truth class and YOLO configuration.

Configuration	Fire images	No-fire images
No YOLO	25/100 (25%)	31/100 (31%)
YOLO	25/100 (25%)	28/100 (28%)

Only 62.25% of Qwen3-VL inferences produced fully valid output across both runs. Hard failures, such as timeouts, empty responses, and invalid structure, account for approximately 27.25% and are visible as error bars in the confusion matrices

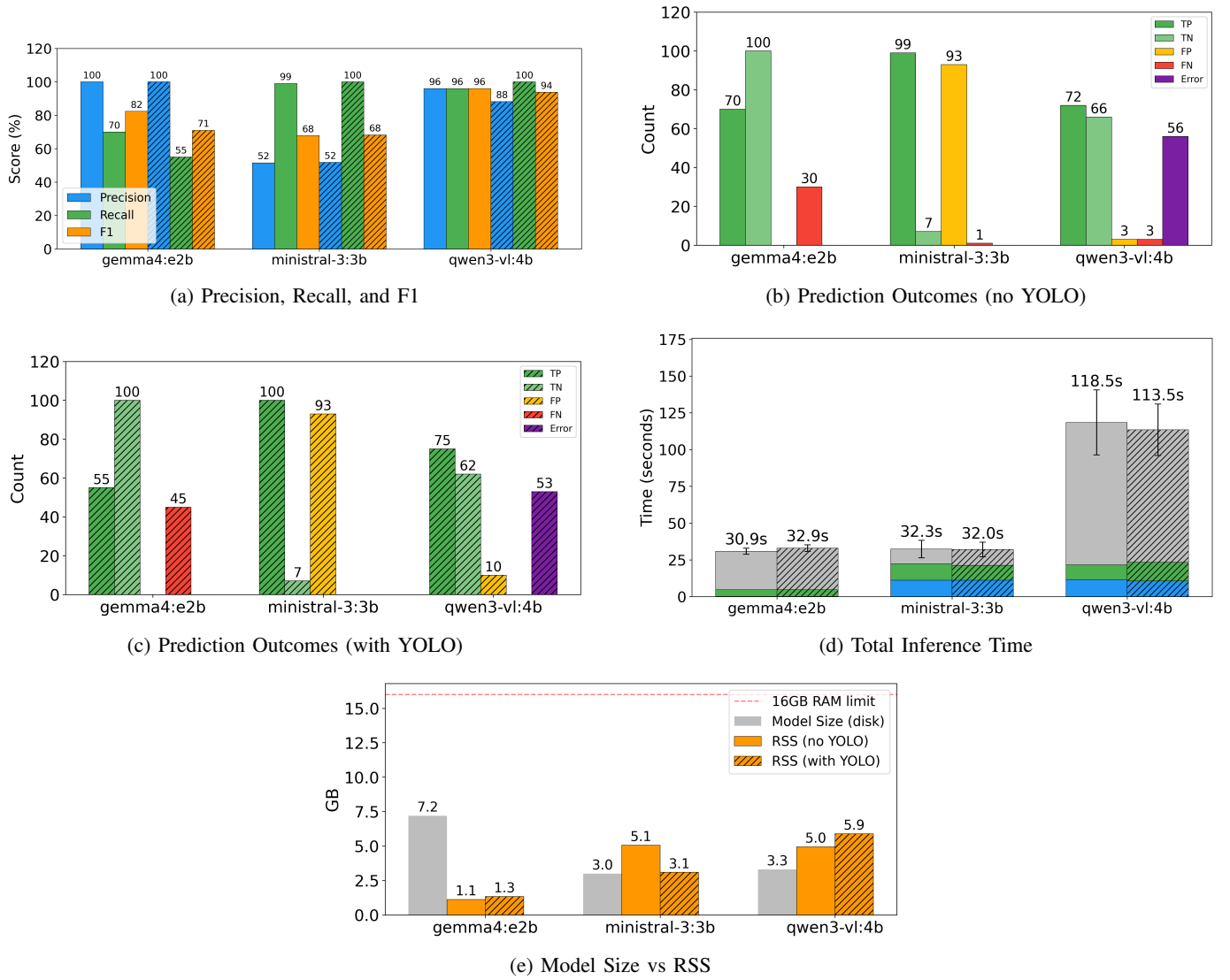


Figure 6: Cycle 2 LLM benchmark results — classification performance and resource utilisation across YOLO and no-YOLO configurations.

(Figures 6b and 6c). The remaining 10.5% produced garbled content where reasoning leaked into output fields. These are notably not reflected in the confusion matrix error bars as they produced a parseable classification value.

4) *Evaluation Interviews:* Four semi-structured evaluation interviews were conducted with wildfire response professionals (P1–P4) following the Cycle 2 evaluation protocol (Appendix H). Participants first assessed the scenario map and image without system output, then reviewed the generated decision support, and finally compared outputs generated with and without YOLO-annotated input images.

a) *Initial scenario assessment:* Fire extent was the most consistently requested piece of missing information. P1 stated that scale was critical to resource decisions. P4 echoed this, noting that in practice a drone would be dispatched immediately to estimate extent, and that an aerial image of the kind shown could already support a rough estimate if scale information were available. P2 and P3 both identified access

routes, water sources compatible with both ground pumps and helicopters, and wind direction as their primary information needs, in that order.

P2 and P3 raised a specific operational constraint absent from the system: helicopters are prohibited from drawing water from certain bodies, and the relevant restrictions are held in county authority (*länsstyrelsen*) GIS layers. Knowing whether a nearby lake was suitable for helicopter scooping versus motor pump use was described as information that otherwise takes time to confirm and can delay operations. P2 also noted that the road access data lacked load-bearing capacity information, which is a recurring practical problem on forest tracks.

P2 introduced the SITAC standard (*Samverkanskartan*), a Swedish interoperability standard defining map symbology used across all rescue services, and noted that the system’s map symbology was not aligned with it. P2 further noted

that scale rings at 500m and 1km from the fire origin would help responders plan containment lines without manual measurement, and suggested that a spread rate in meters per hour would allow this to be cross-referenced with elapsed time.

b) Decision support usefulness: P1 noted that several elements directly matched their pre-output assessment: the recommendation of aerial resources, the identification of a named water source, the characterization of the approach route, and the suggestion to monitor nearby settlements, describing these as correctly-typed recommendations at the right level of abstraction for a command officer in the initial phase of an incident.

P1 drew a distinction between recommendations that added informational value and those that did not. Generic guidance such as “use available water sources” was described as adding no decision support, since it would be correct under any wild-fire scenario and requires no interpretation. Specific references to named locations, distances, and resource categories were described as more useful because they were verifiable and could be acted on or disagreed with.

P1 and P4 independently identified the absence of fire danger indices (*brandriskindex*, Fire Weather Index) as a notable gap, noting that fuel moisture and drought conditions significantly affect spread rate and therefore the urgency of resource decisions. P4 also noted that forest fuel type or vegetation class was absent, describing this as a common unknown on arrival at a forest fire scene. P1 suggested replacing the numeric slope value with a categorical description such as “steep” or “flat,” as the percentage figure would not be readily interpretable by most field personnel.

P1 noted that the appropriate recommendation level for a command officer was whether additional or aerial resources should be requested, not which station’s crew should respond.

P2 and P3 noted that key elements of the decision support such as approach routes and water sources, would be more useful if reflected as annotations on the map rather than appearing only in the text panel, and both cited SITAC symbol compatibility as the mechanism through which this should be implemented.

c) Presentation and intelligibility: The text volume was described as appropriate by all four participants. P1 and P2 described the bulleted recommendation section as easy to scan. P1 suggested that icons or directional symbols could improve scannability further, while P4 asked for greater visual differentiation between sections through typographic emphasis rather than uniform body text.

Generation time was described as acceptable by all four participants. P1 noted that wildfire incidents do not demand sub-second responses, and that the observed latency was within the range that a responding officer would find usable during travel to the scene.

d) YOLO annotation comparison: All four participants preferred the output without YOLO bounding box confidence scores (Output B). P1 stated that a value like 0.6 is unintelligible without a reference distribution and would require prior training to interpret meaningfully, and that qualitative smoke

description (density, color, and behavior) is more operationally useful. P2 and P3 noted that visible smoke will be visible to the operator regardless, and that numerical scores would add noise rather than clarity. P2 further suggested the scores would risk confusion with established indices such as the FWI, which already serves as the appropriate quantitative layer for fire danger assessment. P4 stated the scores were unnecessary.

e) Comparison with existing systems: None of the four participants identified an existing system directly comparable to the artifact. P1 described current tactical coordination as primarily whiteboard-based, supplemented by digital mapping tools that support resource positioning and fire perimeter marking. P1 noted that such tools serve a different phase of the incident, tracking deployed resources and managing logistics, whereas the artifact addresses the initial assessment and inbound decision phase, for which no equivalent support currently exists. P2 and P3 gave similar assessments.

f) Additional observations and suggestions: P1 mentioned the Copernicus Emergency Management Service (EFFIS) as a potential data source for fire danger indices, noting that the Fire Weather Index is openly available and coordinate-referenced, which would allow the system to query the current danger level for the incident location. and P2 raised that a calculated spread rate derived from wind speed and fire danger index would allow responders to judge how quickly a proposed containment line needed to be reached without manual computation.

5) Model Output Assessment: Scenario 02 was selected for manual inspection of model output quality across all three models, using the same image and context block under both YOLO and no-YOLO configurations. Full outputs are provided in Appendix C.

Ministral 3B produced the most complete context utilisation, referencing named settlements, roads, a named water source with supply category, the nearest fire station with distance, and correctly identifying access limitations in the key constraints field. Output was near-identical with and without YOLO. The same named locations, distances, and tactical priorities appeared in both configurations.

Qwen3-VL 4B referenced all three fire stations with distances, the power line hazard, and the water supply category limitation, which is the most thorough context utilisation of the three models. However, the reasoning field contained recommendation content embedded within it, a partial field boundary violation consistent with the thinking mode leakage described in Section VI-B3. Qwen failed to produce any output on this scenario without YOLO, despite succeeding with YOLO enabled on the identical image and context.

Gemma 4 E2B correctly classified the image and produced relevant key constraints, but its reasoning and recommendation were notably shallow. Despite the context containing five named water sources, three fire stations, and multiple named settlements, the recommendation referenced only a single road.

The garbled output example in Appendix C illustrates the content-garbled failure mode in concrete terms. The model’s internal deliberation appeared in the reasoning field rather than

the reasoning itself, producing 320+ words of output consistent with the upper outliers in Fig. 10d.

VII. DISCUSSION

This study set out to investigate two questions. The first asked how multimodal lightweight LLMs can complement traditional object detection in filtering, interpreting, and explaining visual detection results for tactical decision support (RQ1). The second asked what trade-offs exist between different multimodal LLM architectures in terms of detection accuracy, inference speed, hardware requirements, and cognitive support effectiveness (RQ2).

A. RQ1: Complementing object detectors with LLMs

Detection hierarchy and role separation: Considered independently, the two components reveal a natural processing hierarchy. YOLOv8 achieved 85% image-level classification accuracy at a 52ms inference speed. The LLMs, evaluated without bounding boxes and annotations in Cycle 1, exceeded this, Qwen3-VL at 99%, Ministral at 90%, and Gemma at 82%. This suggests that the LLM is a stronger classifier, but with inference times between 15 and 26 seconds per image, it cannot serve as a first-pass filter. YOLO’s role is therefore more suitable as a preprocessing gate, screening incoming imagery and passing flagged frames to the LLM. In practice however, a naive pass-through of every flagged frame would still be inefficient, YOLO produced 11 false positives out of 50 non-fire images, meaning a significant number of irrelevant frames would reach an expensive inference stage. More fundamentally, a deployed system would not operate on isolated frames but on a continuous stream of imagery from a moving drone, opening the possibility of multi-frame aggregation strategies that could substantially reduce both false positives and false negatives at the preprocessing gate. Regarding the form of input the LLM receives, it has direct implications for classification accuracy. Cycle 2 benchmarks show that YOLO-annotated images produced neutral or worse classification performance across all three models compared to clean images. This strengthens the case for the preprocessing gate: if YOLO annotations offered no classification benefit to the LLM, its role is best confined to filtering, with unmodified frames passed downstream for reasoning. Prior wildfire detection work treats detection output as a useful signal to be consumed downstream [5], but the finding that YOLO annotations degraded LLM classification in two of three models suggests that detection output is not a neutral addition to a vision-language reasoner, it can interfere with the model’s own visual interpretation, rather than supplement it.

Explainability and decision support quality: Where YOLO’s interpretation is confined to detecting and localizing objects it was trained on, the LLM operates at an entirely different level, reasoning over the broader operational situation by incorporating geographic context such as water sources, road access, terrain, and wind. Unlike traditional object detection, which produces a classification and a confidence score with no

accompanying rationale, the LLM reasoning stage produces a traceable evidence chain. The operator can inspect not just what the system concluded, but also the reasoning behind it, expressed in natural language. This is the property that distinguishes the system as a decision support tool, it gives the operator the basis to critically evaluate the output before acting on it. In Cycle 1, however, this explainability is shallow. As demonstrated in Figure 4, the model largely narrates what is already visually labeled in the annotated image, citing bounding box confidence scores to conclude that fire is present is not a meaningful interpretive leap when the annotated image itself already says “fire”, even less so when the fire is visually unambiguous to any observer. This shallowness was directly confirmed by practitioners. When shown output containing confidence scores, interviewees rejected them as decision-relevant information: one stated that without a reference point a value like 0.6 is meaningless, another said that if there is smoke you will see it, adding that numerical scores would only add noise to an already dense operational picture. This aligns with Chen et al.’s finding that AI confidence indicators only improve collaborative task performance when operators can meaningfully interpret them [6], something practitioners in this study explicitly could not do without prior exposure to the metric. The substantive explainability gain comes in Cycle 2, where geographic context introduces information that is not visible in the frame at all. When the reasoning draws on water source proximity, road accessibility, or wind direction to ground a tactical recommendation, the evidence chain becomes non-trivial, it is producing something the operator could not have read directly off the image. This was confirmed in the evaluation interviews: interviewees independently described their initial assessment as a mental checklist covering exactly these factors, suggesting that the LLM’s geographic reasoning aligns with the actual cognitive structure of operational decision-making in ways that object detection alone cannot. Interviewees responded positively to precisely these outputs: named roads relative to wind direction, terrain slope characterization, and specific water source references were highlighted as useful, confirming that contextually grounded natural language reasoning constitutes a qualitatively different form of decision support than annotated bounding boxes or numeric scores.

Specificity as the determinant of value: The distinction between useful and non-useful explainability was drawn explicitly by practitioners. When reviewing the system output, an interviewee noted that generic advice, such as “use available water sources”, added no decision support value because it would be correct in virtually any wildfire scenario and therefore requires no interpretation. By contrast, specific recommendations referencing named attack routes, aerial resources, and identified water sources were received positively. This suggests that the value of the LLM’s reasoning is proportional to its specificity: generic reasoning that cannot be evaluated by the operator functions no differently from no reasoning at all, while geographically grounded reasoning, specific enough to be verified and potentially disagreed with,

is what constitutes meaningful decision support in practice. This is consistent with Senoner et al.’s finding that explanation quality, rather than explanation presence alone, determines whether AI-generated explanations improve task performance [3]. A generic recommendation that holds in every wildfire scenario is formally an explanation but provides no more decision support than no explanation at all.

B. RQ2: Trade-offs between LLM architectures

Detection Accuracy: Classification accuracy alone is an insufficient basis for model evaluation in this context, as accuracy and reliability are separate properties that do not move together. Cycle 1 established a seemingly clear hierarchy, Qwen3-VL at 99%, Ministral at 90%, Gemma at 82%, that Cycle 2 collapsed entirely. Qwen3-VL retained the strongest valid-output accuracy when it produced output, but failed on more than a quarter of all images under the richer Cycle 2 prompt and context conditions, failures attributable to its thinking mode exhausting the generation budget under increased input complexity. The failures were distributed across both fire and non-fire images at similar rates, with no clear pattern linking failure to image content, making it impossible to anticipate or route around failures at the input level. Peak accuracy is meaningless if the model is operationally unusable, in a time-critical decision support context, a model that frequently produces no output is worse than one that is consistently moderate, because no output is not a neutral outcome, and no position in a Cycle 1 hierarchy predicts this. Ministral’s collapse took a different form. Rather than failing to produce output, it produced output that was almost always wrong in the same direction, classifying the overwhelming majority of non-fire images as fire in both configurations. The introduction of geographic context appears to have overridden visual evidence entirely, turning a competent Cycle 1 classifier into one that effectively detects nothing but fire regardless of what the image shows. An image primacy clause added explicitly to counter this was insufficient. Whether this is fully addressable through targeted prompt engineering remains open, but the severity of the bias suggests it is not a marginal sensitivity. Gemma exhibited the opposite effect. Where Ministral over-detected, Gemma became increasingly conservative, particularly in the YOLO configuration, where missed fire detections increased substantially compared to both its Cycle 1 results and its own no-YOLO Cycle 2 performance. The same contextual pressure that pushed Ministral toward indiscriminate fire detection pushed Gemma toward excessive caution. Context bias therefore did not produce a uniform degradation across models, it produced model-dependent failures in opposite directions, none of which were predictable from Cycle 1 results alone. The asymmetry points to a deeper issue: context is not a neutral addition to a classifier. Introducing geographic context degraded two of three models despite no change in model weights, and a dedicated image primacy instruction was insufficient to counter it in Ministral’s case. Whether this is resolvable through model-specific prompt tuning remains an open question, the evaluation used a shared prompt across

all three architectures, and per-model calibration may yield different results. What the data does show is that context-classifier interference is architectural in nature and cannot be assumed away: it must be explicitly accounted for when designing systems that combine visual classification with rich contextual reasoning.

Existing work, including Yao et al.’s evaluation of MiniCPM V [22] and Mai et al.’s survey of multimodal deployment [19], demonstrates that small multimodal LLMs can match larger models on standardized benchmarks, with context integration identified as a core capability. Our findings reveal the boundary condition of both claims: under simple prompts, the evaluated models performed reliably and efficiently, consistent with prior work. However, when context and prompt complexity increased substantially, this broke down in three distinct model-dependent ways. Context overrode visual evidence in Ministral, exhausted the generation budget in Qwen3-VL, and was largely ignored in favor of shallow output in Gemma. Rather than uniformly augmenting capability, context introduced failure modes that were neither predictable from single-modal benchmarks nor consistent across architectures, suggesting that reported edge performance does not generalize across deployment configurations.

Inference Speed: Latency at this parameter scale is determined primarily by whether thinking mode is engaged. Qwen3-VL’s median inference time increased almost five-fold between cycles despite identical hardware and model weights, thinking mode scales with input complexity, and at approximately two minutes per image under production conditions it is operationally disqualifying regardless of classification strength. Gemma and Ministral, without thinking mode, converged to approximately 31–33 seconds per image in Cycle 2, flattening the architectural differences visible in Cycle 1. This figure was validated in practice, interviewees who observed Ministral’s inference time during the scenario evaluation found it acceptable for the type of decision it supports. Model selection for latency therefore cannot be made from parameter count or simple benchmarks alone: it requires evaluation under the full production prompt, and the threshold for acceptable latency depends on the operational workflow rather than an absolute figure. Latency figures reported here are also hardware-specific, informal observations on M4 hardware suggested meaningfully faster inference than the M2 used for controlled runs, meaning reported figures likely represent a conservative estimate.

Model Suitability: All three models appeared to operate within the 16 GB memory ceiling with significant headroom, meaning hardware footprint did not appear to be the binding constraint. This interpretation should be treated with caution for Gemma 4 E2B, however, since its Cycle 2 memory measurement was anomalously low and should not be treated as reliable evidence of runtime memory usage. The binding constraints are therefore reliability and latency. No single model is best suited for all roles within the system. The appropriate choice depends on what the model is asked to do and which failure mode is more acceptable in the operational

context. Gemma 4 E2B presents the most consistent profile for binary classification with valid output on every image, zero false positives across both configurations, and acceptable latency. However, its context utilisation was notably shallow compared to the other two models. Despite receiving rich operational context including named water sources, settlements, fire stations, and a protected area, outputs frequently referenced only a single access route and omitted the majority of available context, limiting its utility as a decision support tool regardless of classification reliability. Gemma’s speed advantage from Cycle 1 also largely disappeared in Cycle 2 as the richer prompt increased its inference time, and even at peak speed the seconds saved per inference are unlikely to be operationally significant in a human-reviewed workflow. In a multipass architecture where classification and reasoning are separated into dedicated stages, Gemma would likely be displaced by Qwen for validation and Ministral for decision support, leaving it without a clear role in the pipeline. Ministral 3B produced the most reliably structured and detailed decision support output of the three models. It consistently followed the bullet point format, referenced named context features, and produced concise situation briefs. Its classification performance under Cycle 2 conditions is however too unreliable for it to serve as a standalone detector. Its role is therefore better understood as a reasoning and communication layer that activates after a detection has already been confirmed, rather than as a classifier in its own right. Whether its context bias is addressable through further prompt iteration remains an open question. Qwen3-VL 4B occupies an uncomfortable middle ground. It is the strongest classifier when it produces output, but its reliability and latency make it unsuitable as a primary component in a time-critical system. Its high failure rate and inference latency make it impractical as a primary component in a time-critical system. Since classification does not require a rich prompt, a more suitable role in a multi-pass architecture is discussed in Section VII-D1

C. Limitations and Threats to Validity

1) *Dataset:* The El-Madafri dataset used for accuracy benchmarking is a classification dataset not designed for LLM evaluation. Not all images are aerial, some depict ground-level scenes that may not be representative of real reconnaissance imagery from drones or aircraft. All evaluation was conducted on static images, while YOLO’s primary strength is real-time video detection, which was not assessed.

2) *Geographic Data Sources:* OSM and SRTM data were used for GIS extraction and terrain analysis due to their open availability. In a real deployment, higher-resolution official data from Lantmäteriet would be preferable, offering more accurate elevation models and more complete road and infrastructure coverage, particularly in rural and remote areas where OSM data is sparse. Results related to terrain reasoning and context completeness should be interpreted with this in mind.

3) *Model Selection:* The evaluation covered three architectures selected for parameter comparability and geographic origin diversity. Model size was guided by Cycle 1 classification

accuracy, where larger variants showed little improvement. Decision support quality under rich prompts and context had not yet been evaluated at that stage. Findings may not generalise to other model families or parameter scales.

As model weights are updated frequently, specific accuracy figures should be interpreted as a snapshot at the time of evaluation rather than definitive benchmarks. The system-level findings are more durable: context-classifier interference, the instability of model rankings across system configurations, and the relationship between output specificity and decision support value are properties of the problem structure rather than of any particular model version.

4) *Benchmarking Conditions:* All benchmarks were run on a single hardware class (MacBook Air M2), and latency figures may not generalize to other Apple Silicon generations or non-Apple hardware. No standardization was applied to background system activity during benchmark runs, which may have introduced variance in timing measurements. In Cycle 2, a single pre-computed scenario was used as geographic context across all 200 accuracy benchmark images, meaning classification results under context conditions reflect one operational scenario only.

5) *Measurement Accuracy:* Memory usage was tracked via process-level RSS, which is an approximation on Apple Silicon’s unified memory architecture and may not capture GPU or Neural Engine allocations separately. Ollama’s timing attribution across prompt evaluation, generation, and overhead buckets may not be consistent across model architectures, making component-level comparisons approximate.

An additional measurement anomaly was observed in the Cycle 2 memory results for Gemma 4 E2B, where reported RSS of 1.1 GB without YOLO and 1.3 GB with YOLO was substantially lower than both the model’s 7.2 GB on-disk size and its Cycle 1 measurement, making these figures implausible as a representation of true runtime memory footprint. This likely reflects limitations in the RSS-based measurement method rather than actual memory usage, possibly due to Ollama’s process structure, memory-mapped model weights, or Apple Silicon unified-memory accounting. The Cycle 2 Gemma memory values should therefore not be treated as reliable evidence of runtime memory requirements. Whether this anomaly also affected inference characteristics more broadly, for example contributing to Gemma’s shallower outputs in Cycle 2, could not be independently verified, and no causal relationship can be concluded from the available data.

6) *Qualitative Evaluation:* Qualitative output assessment was based on a single scenario and may not generalize to other scenario types where context utilization patterns differ. Stakeholder interviews only evaluated decision support output from Ministral, meaning practitioner responses to Gemma or Qwen3-VL output were not collected, leaving cognitive support effectiveness partially unevaluated for two of the three architectures.

7) *Stakeholder Interviews and Benchmark Image Selection:* Four interviews were conducted with wildfire specialists from a limited number of organizations, which may not represent the

full range of operational contexts and practices. Practitioners were not involved in selecting or ranking the benchmark images, which were assembled by balanced random sampling. Domain experts might have prioritised operationally representative cases such as early-stage fires, which are both the hardest detection cases and the ones where automated monitoring adds the most value. The absence of expert-guided selection is therefore a threat to ecological validity.

8) *Difficulty Distribution*: As a reproducible proxy for fire extent, the percentage of image area covered by YOLO fire detections was computed for each of the 100 fire images in the Cycle 2 benchmark (Fig. 11). The measure is deliberately coarse, since bounding boxes overestimate actual burning area, but it provides an objective ranking of images without relying on subjective judgement. Extent is broadly spread, with tertile thresholds at 47.7% and 67.4%, but 18 images (18%) fall at or below 1% coverage and 12 (12%) produce no detection at all. The sample therefore contains hard cases, though it was not weighted toward them by design. Headline classification figures should be read as an upper bound relative to an operationally weighted distribution. The system-level findings do not depend on the difficulty mix of the benchmark. It should also be noted that low or zero coverage may reflect YOLO detection failures rather than genuinely small fires, meaning the measure captures YOLO-visible fire extent rather than true fire extent.

9) *Wind Data*: Wind is mocked per scenario and does not reflect real meteorological data. Evaluation of wind-driven reasoning reflects model behavior under idealized conditions rather than the variability of real weather input.

D. Future Work

1) *Multi-Pass MLLM Pipeline*: The evaluation indicates that no single model is best suited for all tasks in the pipeline. The models showed different strengths depending on whether the task concerned binary fire classification, contextual reasoning, or communication-oriented decision support. Future work should therefore investigate a multi-pass MLLM architecture in which separate models are assigned narrower roles rather than requiring one model to perform the full task end-to-end. One possible implementation would be to use Qwen3-VL 4B as a classification or validation stage, while using Ministral 3B as a reasoning and communication stage after a fire detection has been confirmed. This would reflect the observed strengths of the two models: Qwen3-VL 4B showed stronger binary classification performance under a minimal system prompt, while Ministral 3B generated more detailed and structured decision-support output. Separating these responsibilities could reduce the impact of each model's weaknesses and improve the reliability of the overall system. However, such an architecture would also introduce additional latency and orchestration complexity, which would need to be evaluated under realistic edge-deployment conditions.

2) *Image-Based Environmental Inference*: Future work should evaluate whether edge-deployed MLLMs can infer additional environmental information directly from wildfire

imagery, such as broad vegetation and fuel types from visible characteristics, or local wind influence from smoke plume direction. These capabilities would need careful evaluation before operational use, as forest-type classification from RGB imagery is sensitive to resolution, season, and smoke occlusion, and image-based estimates should be treated as tentative until validated against ground-truth data.

3) *Evaluating Larger Local MLLMs*: This thesis limited the final evaluation to locally deployed MLLMs in the 2–4B effective parameter range, based on preliminary tests where larger variants did not show improvements in classification accuracy or reasoning quality that justified their higher inference cost. However, Cycle 2 findings suggest that model scale may become more important as task complexity increases. When the prompt was expanded to include geographic context, structured output requirements, and detailed tactical reasoning, several reliability issues emerged, most notably Qwen3-VL 4B producing empty, invalid, or garbled outputs on a substantial share of inputs. Performance on a narrow binary classification task may therefore not predict performance under richer operational prompts. Future work should evaluate whether larger locally deployed MLLMs, such as 7–8B variants, handle complex prompts and contextual decision support more reliably, measuring not only classification accuracy but also output validity, schema adherence, context utilisation, hallucination rate, and failure frequency.

VIII. CONCLUSION

Wildfire reconnaissance demands rapid, informed decisions under conditions that preclude reliable connectivity and limit human attention. This thesis addressed that gap by designing, implementing, and evaluating an edge-deployable pipeline that combines object detection with multimodal LLM reasoning to produce natural language decision support grounded in geographic context. This represents, to our knowledge, one of the first demonstrations that this class of system is feasible on consumer-grade laptops without network dependency.

The central finding is not simply that small MLLMs can run at the edge, but that their value as decision support tools is contingent on what they are given to reason over. A model narrating bounding box labels adds no meaningful interpretation beyond what the operator can already see. A model reasoning over named water sources, road accessibility, terrain slope, and wind direction produces output that aligns with the actual cognitive structure of operational assessment, which is something no object detection model can provide by design. Specificity, not mere explainability, is what distinguishes useful decision support from noise.

At the same time, the results expose a failure mode that the existing literature on edge-deployed multimodal models does not address: context is not a neutral addition to a visual classifier. Introducing geographic context degraded classification reliability in two of three architectures in model-dependent and opposing directions, and caused inference times to increase to roughly four times that of the other two models in a third. This third model also failed to produce any output on over

a third of images under production conditions. These effects were not predictable from single-modal benchmarks and did not respond fully to prompt-level mitigations, indicating that context-classifier interference is architectural in nature. This finding has implications beyond wildfire detection: any system that combines visual classification with rich contextual input must treat this interaction as a first-class design concern rather than an implementation detail.

The evaluation is bounded by its hardware platform, dataset, and the small number of stakeholder interviews. The qualitative scenario assessment covered all three models, though only one scenario was used. Stakeholder interviews assessed output from one model only, Ministral 3B. These constraints mean the findings are best understood as establishing feasibility and identifying failure modes rather than providing definitive performance benchmarks for operational deployment.

The results open several productive directions. First, a multi-model architecture can separate classification and reasoning into dedicated stages matched to each model's observed strengths. Additionally, systematic evaluation of larger model variants under complex operational prompts can determine whether the reliability failures observed here are a function of scale or architecture. More broadly, this work suggests that the path toward trustworthy AI decision support in high-stakes environments runs not through more capable models alone, but through system designs that give those models the right information, in the right form, for the right task.

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APPENDIX A

REQUIREMENTS AND TRACEABILITY

A. Cycle 1 Requirements

- **REQ01** — The system shall perform image quality screening before passing imagery to the detection stage.
- **REQ02** — The system shall identify the presence and location of fire and smoke in input imagery.
- **REQ03** — The system shall communicate uncertainty levels alongside its recommendations.
- **REQ04** — The system shall measure and report inference speed metrics for each LLM architecture, including inference duration and tokens generated per second.
- **REQ05** — The system shall measure and report classification performance for each LLM architecture using accuracy, precision, recall, F1 score, and Matthews correlation coefficient.
- **REQ06** — The system shall measure and report peak runtime memory usage for each LLM architecture.
- **REQ07** — The system shall support two benchmark modes: an accuracy mode optimised for statistically meaningful measurement over a large image set, and a speed mode optimised for fair inference-speed comparison under thermally controlled conditions.
- **REQ08** — The system shall support reproducible benchmarking of evaluated models.
- **REQ09** — The system shall support modular integration of different LLM architectures without requiring changes to the detection pipeline.
- **REQ10** — The system shall be evaluated across all LLM architectures on identical hardware configurations to enable fair comparison.
- **REQ11** — The system shall operate fully without cloud connectivity or network access.
- **REQ12** — The system shall run on consumer-grade edge hardware.
- **REQ13** — The system shall generate natural language explanations of its reasoning, not only its recommendations.

B. Cycle 2 Requirements

- **REQ14** — The system shall produce tactical recommendations that integrate fire detection results with geographic context describing the surrounding terrain, infrastructure, and environmental conditions.
- **REQ15** — The system shall indicate the tactical priority of its recommendations.
- **REQ16** — The system shall produce structured output containing the components needed by responders, including a classification, reasoning, recommendation, and a concise communication-ready situation brief.
- **REQ17** — The system shall produce output within a timeframe that does not delay operational decisions.
- **REQ18** — The system shall produce useful output without requiring any mandatory configuration beyond selecting an image or scenario.
- **REQ19** — The system shall operate using only data pre-loaded onto the device, without requiring network connectivity at runtime.
- **REQ20** — The system shall display the operational deployment area on a map alongside the LLM output.

TABLE V: Requirements traceability for Cycle 1 and Cycle 2.

Req.	Component	Status
REQ01	Stage 1	Met. Four OpenCV-based checks.
REQ02	Stage 2	Met. Fine-tuned YOLOv8.
REQ03	Stage 2	Partially met. Uncertainty conveyed visually via bounding boxes; LLM does not produce confidence estimates.
REQ04	Benchmarking	Met. Inference duration and tokens/sec logged per inference.
REQ05	Benchmarking	Met. Accuracy, precision, recall, F1, MCC computed per model.
REQ06	Benchmarking	Met. Peak RSS tracked across Ollama processes.
REQ07	Benchmarking	Met. Accuracy and performance modes implemented.
REQ08	Benchmarking	Met. Fixed seed, shared prompt, shared image sample.
REQ09	Stage 3	Met. Ollama runtime supports modular model swapping.
REQ10	Benchmarking	Met. All runs on identical hardware class.
REQ11	Stage 3	Met. Ollama serves models locally.
REQ12	Artifact	Met. Runs on Apple Silicon consumer hardware.
REQ13	Stage 3	Met. Structured JSON includes natural-language reasoning.
REQ14	Stage 3	Met. OSM and SRTM data extracted offline and passed as structured context to the LLM. Water, roads, settlements, terrain, assets, and wind covered.
REQ15	Stage 3	Descoped. Tactical priority prototyped but found too volatile; removed from schema.
REQ16	Stage 3	Met. Structured output includes classification, reasoning, recommendation, and situation brief.
REQ17	Stage 3	Partially met. Ministral and Gemma within acceptable latency. Qwen3-VL exhibits context exhaustion failures.
REQ18	Artifact	Met. No mandatory configuration required beyond selecting an image or scenario.
REQ19	Stage 3	Met. OSM and SRTM data pre-loaded; no network access required at runtime.
REQ20	UI	Met. Map view implemented showing scenario coordinates as points of interest.

APPENDIX B

USER STORIES

This appendix lists the user stories associated with each system requirement. Stories are grouped by requirement and labelled to match the requirement identifiers used throughout the thesis. Each story follows the format: *As a [role], I want [capability] so that [rationale]*.

A. Cycle 1 Requirements

REQ01 — Image Quality Screening:

- **US01.1** — As a field operator, I want the system to filter out low-quality images so that I can trust the reliability of the detection results.

REQ02 — Fire and Smoke Identification:

- **US02.1** — As a field operator, I want the system to identify the presence and location of fire and smoke in imagery so that I can quickly assess the situation without manually analysing every image.

REQ03 — Uncertainty Communication:

- **US03.1** — As a field operator, I want the system to indicate how confident it is in its detections so that I can assess whether the imagery analysis is reliable.

REQ04 — Inference Speed Metrics:

- **US04.1** — As a researcher, I want to measure and report inference duration for each compared architecture so that I can evaluate whether there is a meaningful difference in responsiveness between models.
- **US04.2** — As a researcher, I want to measure tokens generated per second so that I can compare generation throughput independently of total response length.

REQ05 — Classification Performance Metrics:

- **US05.1** — As a researcher, I want to measure and report fire detection accuracy for each LLM architecture so that I can evaluate whether there is a meaningful difference in accuracy between models.
- **US05.2** — As a researcher, I want to measure precision and recall so that I can distinguish between models that over-predict fire and models that miss real fires.
- **US05.3** — As a researcher, I want to report F1 and Matthews correlation coefficient so that I have metrics that remain informative on imbalanced class distributions.

REQ06 — Peak Runtime Memory:

- **US06.1** — As a researcher, I want to measure and report peak runtime memory for each LLM architecture so that I can evaluate whether each model is viable on consumer-grade edge hardware.

REQ07 — Benchmark Modes:

- **US07.1** — As a researcher, I want a dedicated accuracy benchmark mode so that classification metrics are computed over a sufficiently large image set to be statistically meaningful.
- **US07.2** — As a researcher, I want a dedicated speed benchmark mode so that inference-speed comparisons between models are not distorted by thermal throttling or background load.

REQ08 — Reproducible Benchmarks:

- **US08.1** — As a researcher, I want benchmark results to be reproducible so that model evaluations can be verified and compared with confidence.

REQ09 — Modular LLM Integration:

- **US09.1** — As a researcher, I want to swap LLM architectures without modifying the detection pipeline so that I can benchmark different models under identical conditions.

REQ10 — Identical Hardware Evaluation:

- **US10.1** — As a researcher, I want all LLM architectures to be evaluated on identical hardware so that performance differences can be attributed to the models rather than the environment.

REQ11 — Offline Operation:

- **US11.1** — As a field operator, I want the system to run entirely offline so that I can rely on it in areas with limited or no network coverage.

REQ12 — Consumer-Grade Hardware:

- **US12.1** — As a field operator, I want the system to run on a standard laptop so that I can deploy it in the field without specialised equipment.

REQ13 — Natural Language Explanations:

- **US13.1** — As a field operator, I want the system to explain its reasoning in natural language so that I can assess whether its recommendations are trustworthy before acting on them.

B. Cycle 2 Requirements

REQ14 — Geographic Context Integration:

- **US14.1** — As a field operator, I want the system to combine detection results with contextual data so that recommendations reflect the full operational picture rather than imagery alone.
- **US14.2** — As a field operator, I want the system to identify the nearest water source so that I can plan water supply logistics.
- **US14.3** — As a field operator, I want the system to factor in nearby roads so that recommendations account for accessibility for ground crews.
- **US14.4** — As a field operator, I want the system to factor in nearby settlements so that recommendations reflect risk to nearby populations.
- **US14.5** — As a field operator, I want the system to take forest and vegetation type into account so that tactical recommendations reflect the fire behaviour characteristics of the area.
- **US14.6** — As a field operator, I want the system to factor in terrain topology so that recommendations account for slope and accessibility.
- **US14.7** — As a field operator, I want the system to factor in wind direction and speed so that recommendations anticipate likely fire spread.

REQ15 — Tactical Priority (descoped):

- ~~**US15.1** — As a field operator, I want the system to indicate which actions should be prioritised so that I can focus effort on what matters most under time pressure.~~
- ~~**US15.2** — As a field operator, I want the system to flag when civilian evacuation should be prioritised over suppression so that I can shift focus to safety when the situation calls for it.~~

REQ16 — Structured Output:

- **US16.1** — As a field operator, I want the system's output to be structured so that I can quickly locate the information I need without parsing free-form text.
- **US16.2** — As a field operator, I want a concise situation brief so that I can communicate the state of the incident to command or supporting units without manually summarising it.

REQ17 — Response Latency:

- **US17.1** — As a field operator, I want the system's output to be produced quickly enough that it does not delay my first response decisions.

REQ18 — No Mandatory Configuration:

- **US18.1** — As a field operator, I want the system to produce useful output without requiring mandatory configuration beyond selecting an image or scenario, so that I can get situational insight quickly without being slowed down by configuration.

REQ19 — Pre-Loaded Data:

- **US19.1** — As a field operator, I want the system to use only pre-loaded local data so that I can rely on it in environments without network coverage.

REQ20 — Map Display:

- **US20.1** — As a field operator, I want to see the deployment area on a map so that I can orient myself to the geographic context when reviewing the system's recommendations.

APPENDIX C EXAMPLE SYSTEM OUTPUTS

Full model outputs for Scenario 02 (Tyresta NP edge, Stockholm).

A. *Ministral 3B — With YOLO*

Classification: fire_detected

Reasoning: From the image, there is a visible plume of smoke with a density score of 0.78, indicating active fire presence in the forest. The smoke is substantial and appears to be rising vertically, suggesting significant fire intensity. From the context, the terrain is NW-facing with downslope spread potential due to E wind direction, which could exacerbate fire progression toward settlements and high-density buildings. The proximity of numerous buildings and the absence of immediate protected area restrictions highlight the urgency of containment efforts.

Recommendation:

- Deploy suppression teams via the primary service road (0.4 km SW) to secure the fire perimeter near the high-density buildings in Tyresta by and Lillgarden.
- Evacuate all buildings within 2.0 km radius, prioritizing Tyresta by and Svartbacken due to their proximity and high building density.
- Monitor the track road (0.1 km SE) for potential fire spread toward Forsen and use the light-supply stream Bylsjobacken (0.2 km E) for water supply.
- Coordinate with the nearest fire station (4.4 km W) for additional resources and support.

Situation brief: Fire detected with active smoke plume in Haninge kommun; immediate evacuation required for high-density buildings near Tyresta by and Lillgarden. Wind from E to W risks downslope spread; prioritize containment via primary service road and track road (0.1 km SE).

Key constraints:

- High-density buildings within 2.0 km -- evacuation required
- Single primary service road (0.4 km SW) for access
- No confirmed accessible track roads for alternative deployment

B. Ministral 3B — Without YOLO

Classification: fire_detected

Reasoning: From the image, there is a visible plume of smoke rising from a forested area, indicating an active fire. The dense, turbulent smoke suggests significant fire intensity and potential rapid spread. From the context, the terrain is NW-facing with downslope spread risk due to E wind, and the area contains high-density buildings within 2.0 km that require immediate evacuation consideration.

Recommendation:

- Deploy suppression teams via the primary service road (0.4 km SW) to secure the fire perimeter.
- Evacuate all buildings in Tyresta by and Lillgarden immediately.
- Monitor stream Bylsjobacken (0.2 km E) for water supply.
- Establish a perimeter around Forsen (2.3 km N) to prevent spread.
- Coordinate with the nearest fire station in Svartbacken (2.3 km W) for additional resources.

Situation brief: Fire detected; immediate evacuation required for high-density buildings in Tyresta by. Downslope spread risk due to E wind; prioritize suppression via primary service road.

Key constraints:

- High-density buildings within 2.0 km requiring evacuation
- Track road 0.2 km S not confirmed accessible
- Downslope spread risk on NW-facing terrain

C. Qwen3-VL 4B — With YOLO

Classification: fire_detected

Reasoning: From the image, a large, dense smoke plume is visible, indicating an active fire. The context describes a forested, NW-facing slope with a high-density settlement (Tyresta by) 0.7km south and power lines present. Key constraints include: high-density settlement requiring immediate evacuation, power lines posing a safety risk, and limited water supply capacity.

Recommendation: Evacuate residents of Tyresta by (0.7km S) immediately; deploy to service road (0.4km SW) for rapid access; utilize stream Bylsjobacken (0.2km E) for water supply; monitor power lines; coordinate with fire stations (4.4km W, 7.9km N, 10.7km E) for resource deployment.

Recommendation:

- Evacuate residents of Tyresta by (0.7km S) immediately.
- Deploy to service road (0.4km SW) for rapid access.
- Utilize stream Bylsjobacken (0.2km E) for water supply, though light supply.
- Monitor for power line safety hazards.
- Coordinate with fire stations (4.4km W, 7.9km N, 10.7km E).

Situation brief: Active fire with heavy smoke detected; immediate evacuation of Tyresta by (0.7km S) required; water sources available but light supply.

Key constraints:

- High-density settlement Tyresta by 0.7km S -- evacuation priority
- Power lines present -- safety risk
- Light supply water sources -- limited capacity

D. Qwen3-VL 4B — Without YOLO

Qwen3-VL 4B failed to produce any output on this scenario without YOLO preprocessing, returning an empty response. This is consistent with the inference reliability findings reported in Sections VI-B3 and VI-B5.

E. Gemma 4 E2B — With YOLO

Classification: fire_detected

Reasoning: From the image, significant smoke is visible across the landscape, indicating an active fire or heavy smoke presence. From the context, the area is characterized by forest land with a gentle slope and several nearby water sources and settlements.

Recommendation:

- Deploy resources to the area near the visible smoke, prioritizing access via the service road 0.4 km SW.

Situation brief: Active smoke is visible in the forested area; immediate assessment of surrounding settlements and water access is required.

Key constraints:

- Road access limited to service road (0.4 km SW)
- High density buildings (83) require immediate evacuation consideration
- Downslope spread risk due to E wind direction

F. Qwen3-VL 4B — Garbled Output Example

The following output was produced during the accuracy benchmark on a no-fire image. The model classified correctly but leaked internal deliberation into the reasoning field. This example predates the addition of the *key_constraints* field.

Classification: no_fire_detected

Reasoning: From the image, there is no visible fire, smoke, or flames; the scene depicts a foggy autumn landscape with a river and forested. Wait, no, the user's message is correct. Let's make sure the reasoning is accurate. The image has no fire, so classification is no_fire_detected. [...] Let's check the bullet points: each starts with a verb, references context. Yes. So the final output should be as structured above, with the correct fields filled in. The key is to not mix up the classification with the context, and to attribute correctly...

Situation brief: No fire detected; monitoring for fire activity in forested terrain with water sources available; prepare for rapid deployment to nearest water source and settlements.

APPENDIX D PROJECT TIMELINE

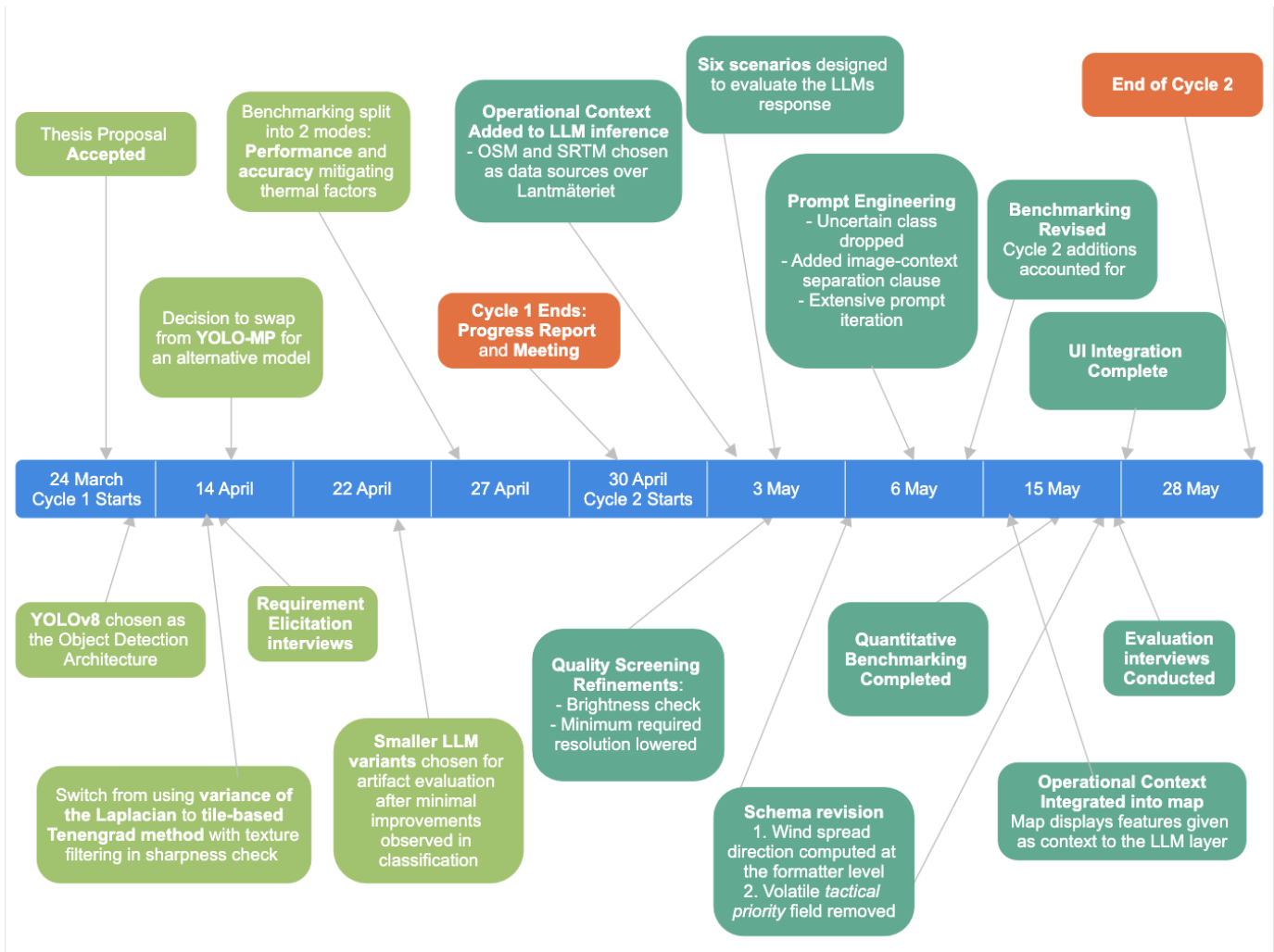


Fig. 7: Timeline Over Major Events and Architectural Decisions

APPENDIX E YOLO DETECTION PERFORMANCE

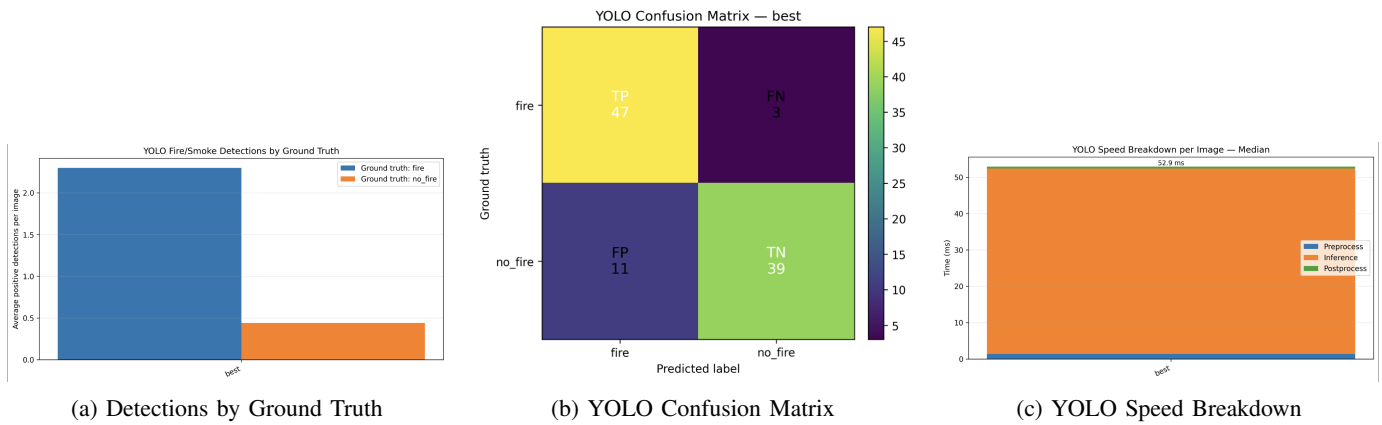


Fig. 8: YOLO Wildfire Detection Performance: (a) detection counts; (b) confusion matrix; (c) latency.

APPENDIX F CYCLE 1 ADDITIONAL BENCHMARK CHARTS

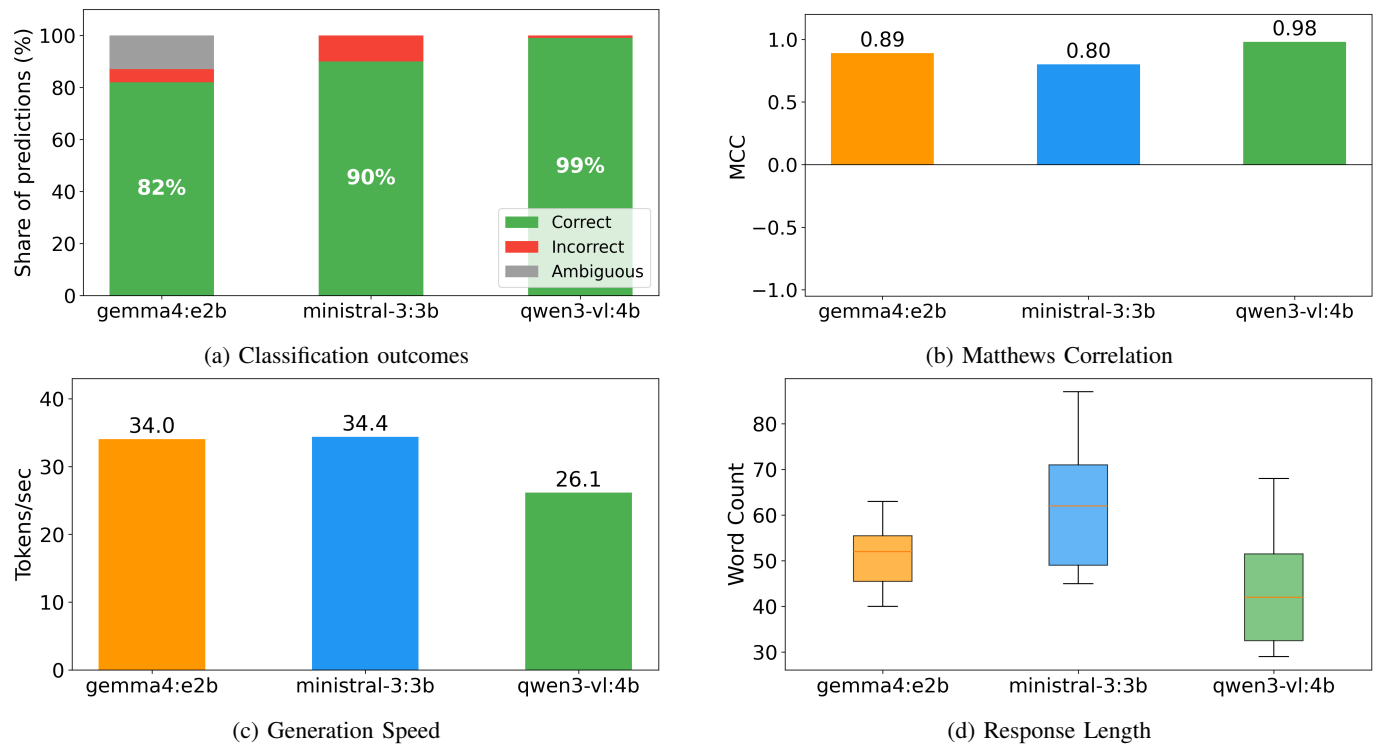


Fig. 9: Cycle 1 additional benchmark results: classification outcomes, Matthews correlation, generation speed, and response length.

APPENDIX G CYCLE 2 ADDITIONAL BENCHMARK CHARTS

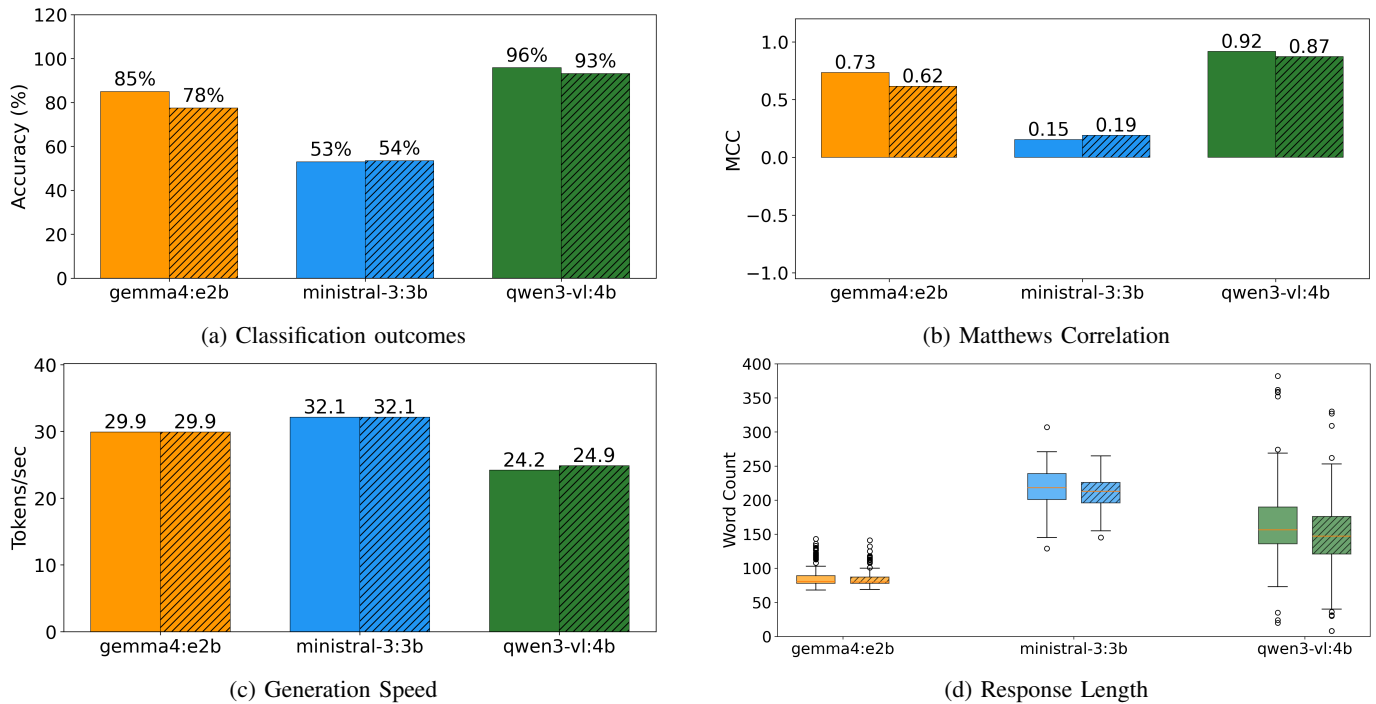


Fig. 10: Cycle 2 additional benchmark results: classification outcomes, Matthews correlation, generation speed, and response length.

APPENDIX H INTERVIEW PROTOCOL

A. Cycle 1: Requirement Elicitation

Semi-structured, audio-recorded with informed consent. Estimated duration: 30 minutes. Questions are in Swedish.

Introduktion

Tack för att du tar dig tid att delta i denna intervju. Syftet med studien är att förstå hur arbete vid skogsbrandsinsatser genomförs i praktiken, med särskilt fokus på hur beslut fattas i verkliga situationer.

Ditt deltagande är helt frivilligt. Du kan när som helst avbryta intervjun eller välja att inte svara på en enskild fråga, utan att behöva förklara varför. Studien följer dataskyddsförordningen (GDPR). All data behandlas konfidentiellt och används enbart för forskningsändamål. Dina uppgifter anonymiseras och kan inte kopplas till dig som individ.

Vi skulle vilja spela in intervjun för att möjliggöra transkribering och säkerställa att inga viktiga detaljer missas.

- Godkänner du att vi spelar in?
- Har du några frågor innan vi börjar?

Projektbeskrivning

Detta kandidatarbete undersöker utvecklingen av ett beslutsstödsystem där språkmodeller assisterar traditionella objektkänningssystem vid identifiering av skogsbränder från flygbilder, samt stödjer beslutsfattande i ett tidigt skede utifrån kontextuell data.

Del 1 — Bakgrund och roll

- Kan du kortfattat beskriva din nuvarande roll och dina ansvarsområden?
- Hur ser en typisk insats ut för dig?

Del 2 — Genomgång av en nylig insats

Vi skulle vilja att du beskriver en skogsbrandsinsats du medverkat i, från det att du först fick kännedom om den tills situationen var under kontroll.

- Hur uppstod händelsen och hur blev du informerad?
- Vad var dina initiala åtgärder när du fick larmet?
- Hur utvecklades situationen därefter?

Del 3 — Beslutspunkter

- Vilken information skulle du beskriva som mest kritisk under insatsens inledande fas — både på väg till platsen och vid ankomst?
- Vilka typer av beslut behöver typiskt fattas under denna fas?

Del 4 — Upptäckt och initial bedömning

- Hur upptäcks skogsbränder vanligen enligt din erfarenhet?
- Används flygbilder i verksamheten idag — exempelvis från flygplan, drönare eller satellit?
- Vilken typ av information är mest värdefull att utvinna från bilder och geografisk kontext i ett tidigt skede?

Del 5 — Reaktion på ett framtida beslutsstödsystem

- Om det hade funnits ett system som kunde hjälpa dig och dina kollegor att tolka information och stödja beslutsfattande — vad skulle du efterfråga av ett sådant system avseende funktionalitet och datainput?

Del 6 — Avslutning

- Finns det något relevant som vi inte har berört?

B. Cycle 2: Evaluation

Semi-structured, audio-recorded with informed consent. Questions are in Swedish.

Introduktion

Vi gör ett kandidatarbete på Göteborgs universitet i samarbete med Försvarets Materielverk. I samband med detta har vi utvecklat en mjukvaruprototyp för beslutsstöd i de initiala stadierna av skogsbrandsbekämpning. Mjukvaran kan köras helt offline, utan internettillgång.

Det finns inga rätt eller fel svar och alla synpunkter är av värde. För att undvika partiskhet kommer vi inte berätta mer om hur systemet fungerar innan intervjun är klar.

Del 1 — Scenariogenomgång

Ladda scenario, visa karta och bild. Flytta runt och zooma ut kartan.

- Utifrån vad du blir presenterad i systemet, vilken typ av information eftersöker du, vilka bedömningar kan du göra och vilka beslut kan du fatta?
- Vad är det första du hade velat göra?

Del 2 — Utvärdering av genererat beslutsstöd

Generera output

- Vad har du att säga om hur lång tid genereringen tog?
- Om du läser beslutsstödet, vad är dina tankar om dess användbarhet?
- Vad tycker du om presentationen av beslutsstödet?
- Vad tycker du om hur förståeligt beslutsstödet är?
- Kan detta system jämföras med något system du använder eller har använt?
- Utöver det du ser, vad tycker du saknas i ett system som detta?

Del 3 — Jämförelse mellan två beslutsstöd

Visa Output A (med YOLO-annoteringar) och Output B (utan YOLO-annoteringar) för samma scenario.

Skillnaden är att Output A genererades med en internt annoterad bild med objektigenkänning för skogsbrand, och Output B med en vanlig bild som input.

- Vilken output föredrar du och varför?

Del 4 — Avslutning

- Finns det något du vill tillägga som vi inte har berört?

APPENDIX I IMAGE SUBSET

The following is the image subset, from the El-Madafiri wildfire dataset [26], that was used for benchmarking. In Cycle 1 the first 100 were used, and in Cycle 2, all 200.

```
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2. test/fire/33116128054_0169d12131_o.jpg
3. test/fire/34549858860_4a9183e9af_o.jpg
4. test/fire/34957427404_14d39d5c12_o.jpg
5. test/fire/36204002431_91b77a7dc5_o.jpg
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APPENDIX J MODEL DIGEST HASHES

The following SHA-256 digests identify the exact model weights used in all benchmarks, corresponding to the application/vnd.ollama.image.model layer in each Ollama manifest.

- **Ministral 3B** (ministral-3:3b):
910e4bf4e2338f181e99796d7452404e85c1b6bbbfb8cd0bb094672cf9b6f2f22
- **Qwen3-VL 4B** (qwen3-vl:4b):
9c60bdd691c1897bbfe5ddbc67336848e18c346b7ee2ab8541b135f208e5bb38
- **Gemma 4 E2B** (gemma4:e2b):
4e30e2665218745ef463f722c0bf86be0cab6ee676320f1cfad91e989107448
- **YOLOv8** (fire_s.pt):
44d51ac7e8e6118c67a7dc974c1b0d869c5b252ee8690dd1c561c2b409b8576b

APPENDIX K
DISTRIBUTION OF FIRE EXTENT

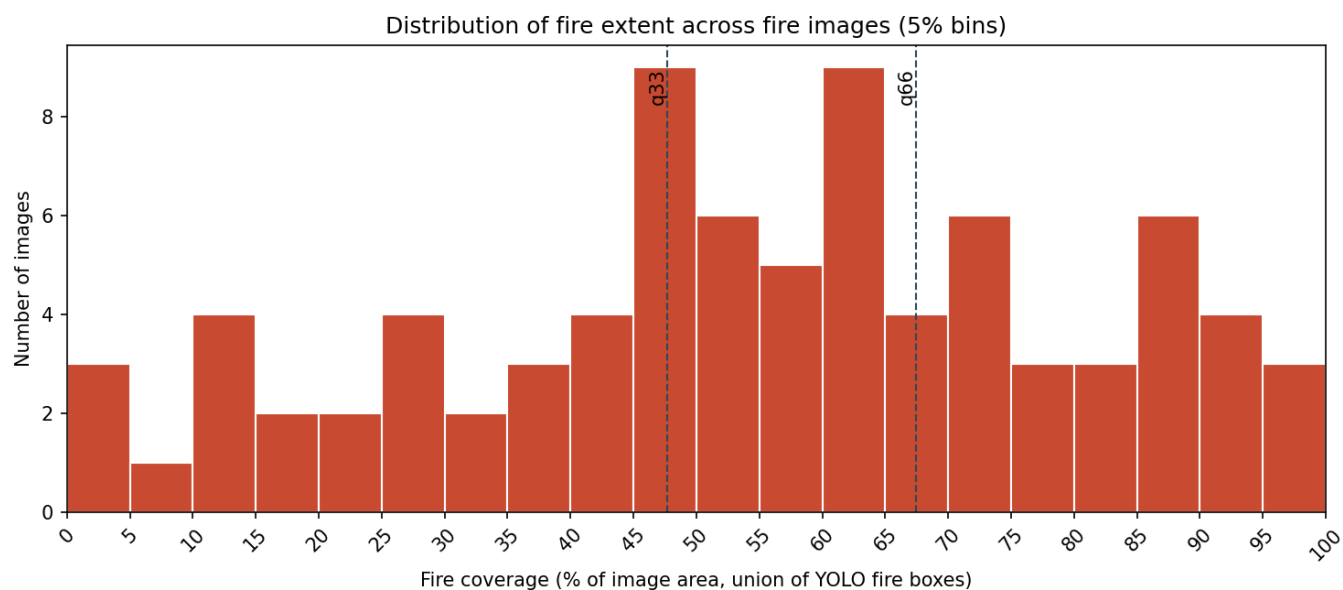


Fig. 11: Distribution of fire extent across the 100 fire images in the Cycle 2 benchmark, measured as the percentage of image area covered by the union of YOLO fire detections (5% bins). Dashed lines mark the tertile thresholds used for difficulty binning ($q_{33} = 47.7\%$, $q_{66} = 67.4\%$).